

HSH

HEAVY SAFE HELICOPTER

Generation 1 Master Engineering & IP Diligence Baseline

Document HSH-00 | Revision 26 — Integrated Professional Master Edition

33.0 m Main Rotor / 28,000 kg Planning Configuration / Expanded 21-Feature Survival Architecture

CONCEPTUAL ENGINEERING BASELINE — PRE-SIMULATION / PRE-HARDWARE / PRE-PROTOTYPE

Proprietary Concept Development Material — Not for Public Disclosure

Founder	Amirbahador Atai
Document Number	HSH-00
Revision	Rev 26 — Integrated Professional Master Edition (33.0 m Rotor / 28,000 kg)
Status	FINAL GENERATION 1 CONCEPTUAL MASTER BASELINE — CONFIGURATION-UPDATED + 21-FEATURE EXPANSION
Development Stage	Pre-Simulation / Pre-Hardware / Pre-Prototype
Intended Audience	Technical due-diligence teams, prospective IP licensees, investors, regulatory liaison
Certification Note	Not a certification approval. No design herein is certified or approved by any authority.
Patent Note	Not a filed patent. IP-A through IP-D are conceptual claim surfaces pending prior-art search.

Engineering Integrity Statement. Every numerical and functional value is a TARGET, ASSUMPTION, STUDY, CALC, CONCEPT or OPEN item unless explicitly marked otherwise. The 28,000 kg figure is TARGET/OPEN. Bottom-up mass closure remains mandatory. AI/ML remains advisory at the safety boundary; the deterministic Safety Kernel is the final authorization authority. Human emergency authority is preserved and non-bypassable. Main-rotor separation is excluded from the core safety case.

Controlling Configuration — Rev 26

Parameter	Value	Status
Design mass	28,000 kg	TARGET / OPEN
Main rotor diameter	33.0 m	TARGET / OPEN
Rotor disk area	855.30 m ²	CALC
Disk loading	32.74 kg/m ²	CALC
Ideal hover power	≈3.14 MW	CALC / STUDY
Shaft power @ FM 0.75	≈4.19 MW	CALC / STUDY
Main rotor / hub / controls	≈3,680 kg	ASSUMPTION / STUDY
F01–F18 physical midpoint	≈1,241 kg	PLANNING / OPEN
Combined midpoint increment	≈+2,021 kg	OPEN — closure required
Occupancy	8 persons maximum	LOCKED LIMIT
Capsule mass allocation	7,000 kg	PLANNING / OPEN
Main-rotor separation	Excluded from core safety case	DEFERRED — RESEARCH

Reading note. The following pages contain the complete controlled engineering baseline (all parts, tables, open items, safety architecture, IP surfaces and verification framework). Where any figure for rotor diameter or design mass differs from the controlling values above (33.0 m / 28,000 kg), the values in this front matter and in PART X (21-feature change set at the end of the document) are controlling. No original engineering text or table has been deleted.

Mass Accounting — 28,000 kg Planning Configuration (Controlling)

Mass Group	Planning Mass	Status
Survival capsule, complete	7,000 kg	TARGET / PLANNING
Main airframe excluding capsule	5,000 kg	ASSUMPTION/STUDY
Main rotor, hub and controls	≈3,680 kg	ASSUMPTION/STUDY
Tail rotor, gearbox and drive	480 kg	ASSUMPTION/STUDY
Engines, transmission, gearboxes and mounts	3,100 kg	ASSUMPTION/STUDY
Fuel system + full usable fuel (FULL)	3,400 kg	ASSUMPTION/STUDY
Rapid separation system	1,000 kg	ASSUMPTION/STUDY
Avionics, electrical, hydraulic, thermal, control	1,600 kg	ASSUMPTION/STUDY
Operating fluids and consumables	320 kg	ASSUMPTION/STUDY
F01–F18 physical survival/recovery features (midpoint)	≈1,241 kg	PLANNING / OPEN (range 852–1,630 kg)
Growth / configuration reserve (adjusted)	≈1,179 kg	PROGRAM RESERVE (indicative)
TOTAL (indicative planning)	≈28,000 kg	TARGET / OPEN — bottom-up closure required

The 28,000 kg figure is a controlled planning target, not a closed MTOW. Midpoint incremental programme mass ≈2,021 kg. F19–F21 carry zero aircraft hardware mass.

Capsule Component Mass — 7,000 kg (Planning Allocation)

Capsule Component (summary groups)	Planning Mass	Notes
Primary crashworthy cell / CFRP roll-cage + titanium nodes	≈2,400–2,800 kg	STUDY — FEA required
Eight energy-absorbing landing supports (interface + absorbers)	≈800–1,000 kg	STUDY — drop-test closure
Eight crashworthy seats + 5-point restraints	≈480–560 kg	≥150 mm stroke; ≤6,670 N
16-airbag occupant protection system	200 kg	TARGET — OI-18 gas analysis open
Life support, emergency power, fire suppression	≈350–450 kg	Independent of primary aircraft systems
Recovery system interfaces + canopy attach structure	≈300–400 kg	3+2 architecture + F07 elements
Avionics, sensors, wiring, thermal protection (baseline)	≈400–500 kg	Before F02/F03/F04/F15 increments
Integration margin + growth within capsule allocation	Remainder to 7,000 kg	OPEN — bottom-up mandatory
TOTAL CAPSULE ALLOCATION	7,000 kg	PLANNING / OPEN

8 persons maximum remains a LOCKED LIMIT. Main-rotor separation remains excluded from the core safety case (DEFERRED — RESEARCH). Human emergency authority is non-bypassable. The complete engineering baseline content begins on the next page. The 21 new survival/recovery/verification features (F01–F21) are placed in full at the end of this document.



Generation 1 Master Engineering & IP Diligence Baseline

Document HSH-00 | Revision 25 — Professional Master Edition, 30.0 m Main-Rotor Configuration

(Supersedes Rev 23 rotor/hover/mass content; carries forward Rev 24 configuration change in full Rev 23 structure and depth)

CONCEPTUAL ENGINEERING BASELINE — PRE-SIMULATION / PRE-HARDWARE / PRE-PROTOTYPE

Proprietary Concept Development Material — Not for Public Disclosure

Founder	Amirbahador Atai
Document Number	HSH-00
Revision	Rev 25 — Professional Master Edition (30.0 m Rotor Configuration)
Supersedes	HSH-00 Rev 12 through Rev 24 (all prior presentation, diligence and configuration packages)
Status	FINAL GENERATION 1 CONCEPTUAL MASTER BASELINE — CONFIGURATION-UPDATED
Development Stage	Pre-Simulation / Pre-Hardware / Pre-Prototype
Intended Audience	Technical due-diligence teams, prospective IP licensees, investors, regulatory liaison
Certification Note	Not a certification approval. No design herein is certified or approved by any authority.
Patent Note	Not a filed patent. IP-A through IP-D are conceptual claim surfaces pending prior-art search.

Engineering Integrity Statement. Every numerical and functional value in this baseline is a TARGET, ASSUMPTION, STUDY, CALC, CONCEPT, or OPEN item unless explicitly marked otherwise. None of these values constitutes demonstrated performance, and nothing in this document shall be read as a certification finding, an airworthiness determination, a granted patent claim, or proof of achieved survival probability. Every status label is defined precisely in Part I, Section 5.

Document Control

Field	Value
Document Title	HSH Heavy Safe Helicopter — Generation 1 Master Engineering & IP Diligence Baseline
Document Number	HSH-00
Revision	Rev 25 — Professional Master Edition (30.0 m Rotor Configuration)
Supersedes	HSH-00 Rev 12 through Rev 24 (all prior presentation, diligence and configuration packages)
Status	FINAL G1 CONCEPTUAL MASTER BASELINE — CONFIGURATION-UPDATED
Founder	Amirbahador Atai
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Classification	Proprietary Concept Development Material — Not for Public Disclosure
Certification Note	Not a certification approval
Patent Note	Not a filed patent

■ Editorial and configuration scope of Rev 25. This edition performs two things at once. First, it carries forward, without technical alteration, the full nine-part presentation structure, working table of contents, executive summary, standardized terminology and units, and reviewer-usability formatting introduced in Rev 23. Second, it applies the Rev 24 configuration change — main rotor diameter increased from 26.0 m to 30.0 m, with the associated disk area, disk loading, hover-power study, main-rotor mass line, and growth-reserve line recalculated and carried through every table, cross-reference and mass rollup in this document — so that no page in this baseline mixes 26.0 m and 30.0 m figures. Where Rev 24 was a compact 10-page configuration-change notice, Rev 25 is the fully reconciled master edition: every Rev 23 section, table and open item is retained, and every rotor-dependent number is restated at the 30.0 m configuration. This edition does not add new engineering analysis, new test evidence, or new closure of any open item beyond what Rev 24 supported, and it does not upgrade any TRL, status label, or claim.

Rev	Primary Change	Status
Rev 12	26,000 kg / 26.0 m / 4.0 m baseline; 8-canopy recovery; single 6 s target; 10–12 occupants	Superseded
Rev 13	Main-rotor separation removed from core; staged S0–S5; 3+2 canopies; Risk Register; Prior Art	Superseded
Rev 14–16	Full narrative depth; helium-hybrid trade posture; competitive IP narrowing	Superseded
Rev 17–18	HSH identity unified; 8-person lock; commercial content separated; Maturity Framework	Superseded
Rev 19	16-airbag architecture; seat specification; OI-16 / OI-17 registered	Superseded
Rev 20	Seat stroke ≥ 150 mm / $\leq 6,670$ N; airbag mass 200 kg; OI-18; IP-D; TRL; competitive; geometry; mission	Superseded
Rev 21–22	Diligence §§32–41; portrait uniformity; dense no-blank composition of full content	Superseded as presentation base
Rev 23	Professional master edition: nine-part structure, working TOC, executive summary, standardized tables/units, consolidated glossary, reviewer-usability pass	Superseded as rotor/mass base
Rev 24	Configuration change: main rotor 26.0 m → 30.0 m; disk area, disk loading, hover power, main-rotor mass, and growth reserve recalculated; 10-page change notice format	Superseded as presentation base
Rev 25	Professional master edition rebuilt at full Rev 23 depth and structure with the Rev 24 30.0 m rotor configuration reconciled through every section, table and mass rollup	CURRENT

Baseline Retention Statement. Every requirement, status label, open item (OI-01 through OI-18, plus R24-OI-01 through R24-OI-05), risk register entry, verification gate, interface definition, safety layer, Safety Kernel precondition set, staged

separation timeline, recovery architecture, prior-art posture, and engineering maturity framework carried in Rev 18 through Rev 24 is retained in Rev 25 without technical alteration except for the rotor-diameter-dependent figures explicitly restated below. No open item is presented as closed, and no status label is upgraded, in this edition.

Executive Summary

HSH Heavy Safe Helicopter (Generation 1) is a conceptual architecture for a 26-ton-class single-main-rotor helicopter organized around one idea: an occupied survival capsule that can become the aircraft's final landing boundary in a catastrophic emergency, supported by a deterministic safety-authorization layer, a staged non-rotor separation sequence, an eight-point capsule-integrated impact-attenuation system, and a distributed parachute recovery system. The programme is positioned as a distinct product category — a survivability-first rotorcraft — rather than a payload or cabin-space competitor to existing helicopter models.

What changed in Rev 25. The main rotor diameter is locked at 30.0 m (previously 26.0 m in Rev 23), giving a disk area of 706.86 m² (previously 530.93 m²) and a disk loading of 36.78 kg/m² (previously 48.97 kg/m²) at the unchanged 26,000 kg design mass. Ideal hover power falls to approximately 3.10 MW and shaft power at FM 0.75 to approximately 4.13 MW (previously approximately 3.57 MW and 4.76 MW respectively). The main-rotor system mass planning line rises to 2,900 kg (previously 2,100 kg), funded by reducing the growth/configuration reserve from 2,000 kg to 1,200 kg, so the 26,000 kg design mass total is unchanged and fuel tanks remain FULL at the 3,400 kg fuel-system line. No other element of the architecture is altered by this configuration change.

What this baseline is. HSH-00 Rev 25 is a Generation 1 conceptual master baseline: a single-source document intended to let a technical due-diligence team, prospective IP licensee, or investor evaluate the architecture without reconstructing it from an underlying tree of hundreds of subordinate documents. It defines top-level geometry and mass allocations at the 30.0 m rotor configuration, a six-layer survival architecture, a Safety Kernel precondition set, a staged S0–S5 separation timeline, a 3+2 canopy recovery architecture, an eight-support landing system, a 16-airbag and crashworthy-seat occupant-protection architecture, four candidate IP claims (IP-A through IP-D), a full open-items register (OI-01 through OI-18, plus R24-OI-01 through R24-OI-05), a technology-readiness assessment, a competitive-differentiation table, and an evidence-ladder framework connecting every claim to the test and analysis evidence required before it can be treated as demonstrated.

What this baseline is not. No element of the design has been analyzed by simulation, demonstrated by test, validated in flight, or certified by a regulator. Every numerical target is a planning allocation, not a verified result. The document deliberately excludes main-rotor separation from the core safety case — a design decision made in Rev 13 and retained here — because no credible six-degree-of-freedom analysis or structural evidence exists to support separating a rotor carrying several hundred megajoules of rotational kinetic energy from a descending occupied capsule. This exclusion is treated as a strength of the architecture, not a gap: a safety case should not depend on its least-proven capability.

Dimension	Current Position
Core proposition	Occupied survival capsule as final landing boundary, gated by a deterministic Safety Kernel, supported by staged non-rotor separation, eight-point impact attenuation, and 3+2 canopy recovery.
Rotor configuration	30.0 m main rotor diameter (Rev 25 / Rev 24 locked value); disk area 706.86 m ² ; disk loading 36.78 kg/m ² ; ideal hover power ≈3.10 MW; shaft power @ FM 0.75 ≈4.13 MW.
Maturity	Specified / Studied across all subsystems. TRL 1–4 by subsystem (Part V). No subsystem has reached subsystem testing (E3) or beyond.
Principal residual risks	Stage S3 capsule release has no certified analog; minimum canopy deployment altitude unresolved; 7,000 kg capsule mass unverified bottom-up; sealed-capsule airbag overpressure/toxicity (OI-18) unanalyzed; six-layer independence undemonstrated; 30.0 m rotor dynamics and clearance unanalyzed (R24-OI-02); reduced 1,200 kg growth reserve.
IP posture	Four candidate claims (IP-A–D), strongest as a system-level combination (IP-D). Conceptual only — not filed; professional prior-art search required before any filing or broad non-NDA disclosure.
Locked configuration	8-occupant maximum (Generation 1 hard limit); main-rotor separation excluded from the core safety case; no retractable landing gear; central capsule position; 30.0 m main rotor diameter.
Evidence path	G0 architecture freeze (approaching closure) through G8 certification/qualification, structured against the E0–E5 evidence ladder (Part VII).

■ This executive summary reflects the Rev 25 configuration. It synthesizes, but does not alter, the technical content of Parts I through IX. Where a summary statement and a detailed section appear to differ, the detailed section is controlling.

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PART I

Programme Baseline, Scope and Status Control

Identity, occupancy lock, scope, philosophy, status-label definitions and configuration-change triggers

1. Baseline Freeze and Controlled Identity

1.1 Programme Identity

The programme name is HSH Heavy Safe Helicopter. The controlling document is HSH-00 Revision 25, Founder: Amirbahador Atai. All prior references to HSC-00 or earlier HSH-00 revisions are superseded for configuration control. Rev 25 carries the complete Rev 20/22/23 technical baseline in a reorganized, consistency-checked presentation, updated throughout for the Rev 24 30.0 m main-rotor configuration.

Item	Value	Status
Programme	HSH Heavy Safe Helicopter Generation 1	LOCKED
Document	HSH-00 Rev 25	CURRENT BASELINE
Design mass	26,000 kg	TARGET
Flight architecture	Single main rotor + tail rotor	LOCKED CONCEPT
Main rotor diameter	30.0 m	TARGET — REV24/25
Tail rotor diameter	4.0 m	TARGET
Capsule mass	7,000 kg	TARGET
Occupancy	8 persons maximum	LOCKED LIMIT
Range target	1,000 km	TARGET
Landing gear	8 short thick fixed supports on capsule; no wheeled retractable gear	LOCKED CONCEPT

1.2 Occupancy Lock

■ 8 persons maximum [LOCKED LIMIT]. This is a Generation 1 hard configuration constraint. Any future change requires formal configuration control and re-analysis of geometry, mass, CG, inertia, seating, restraints, airbag count, life support, egress, structural loads, recovery sizing and certification basis.

1.3 What Rev 24/25 Changed Relative to Rev 23

Item	Rev 23	Rev 24/25	Rationale
Main rotor diameter	26.0 m	30.0 m	Trade rotor size for lower disk loading and hover-power margin
Disk area	530.93 m ²	706.86 m ²	CALC from D = 30.0 m
Disk loading	48.97 kg/m ²	36.78 kg/m ²	CALC — 24.9% reduction at constant design mass
Hover P _{ideal}	≈3.57 MW	≈3.10 MW	CALC — actuator disk theory

Item	Rev 23	Rev 24/25	Rationale
Hover P @ FM 0.75	≈4.76 MW	≈4.13 MW	CALC — study value
Main rotor / hub / controls	2,100 kg	2,900 kg	+800 kg — planning uplift for larger blades/hub
Growth / config reserve	2,000 kg	1,200 kg	–800 kg — funds rotor mass uplift at constant 26,000 kg total
Aircraft design mass	26,000 kg	26,000 kg (FULL fuel)	Unchanged total; fuel tanks explicitly FULL
Fuel system + usable fuel	3,400 kg	3,400 kg	Unchanged line; usable fuel ≈2,600 kg / ≈3,250 L
Capsule / safety architecture	As defined	Unchanged	Six-layer architecture, Kernel, S0–S5, OI-01–18 retained

1.4 What Is Retained Unchanged

Every requirement, status label, open item OI-01 through OI-18, risk register entry, verification gate, interface definition, safety layer, Safety Kernel precondition set, staged separation timeline, recovery architecture, prior-art posture, and engineering maturity framework from Rev 18/19/20/22/23 is retained. Rev 25 additionally carries forward the Rev 24 rotor-configuration change and its five configuration-specific open items (R24-OI-01 through R24-OI-05). No structural alteration beyond rotor-dependent figures and their downstream mass/power consequences is made.

2. Purpose and Scope

2.1 Programme Objective

HSH Generation 1 defines the system-level architecture for a 26-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain — not reliance on any single device. As of Rev 25, the main rotor is locked at 30.0 m diameter to reduce disk loading and increase hover-power margin relative to the Rev 23 26.0 m configuration.

2.2 Core Scoping Decision

The survival chain is designed to work assuming the main rotor is never separated from the aircraft. This deliberate decision was introduced in Rev 13 and is retained. No credible trajectory, gyroscopic, or structural analysis yet exists to support main-rotor separation from a descending occupied capsule. Building the core safety case on an unproven capability would undermine confidence in the entire architecture and in all primary IP claims. The larger 30.0 m rotor of Rev 25 carries more rotational kinetic energy than the Rev 23 26.0 m rotor, which reinforces rather than weakens this scoping decision.

2.3 Scope Coverage

This document covers: the aircraft; the occupied survival capsule; the rapid selective separation system (non-rotor components only); safety initiation paths; deterministic safety authorization; stabilization; recovery; impact attenuation; post-event survival; and verification interfaces. It does not cover detailed structural analysis results, certified engine data, qualification test results, airworthiness findings, or IP legal opinions — those are evidence-gate deliverables.

2.4 Design Philosophy

Separation is a controlled emergency mode, authorized only when defined conditions are satisfied. The survival architecture supports multiple outcomes in priority order: (1) conventional flight recovery; (2) autorotation; (3) selective hazard/mass reduction; (4) capsule separation and parachute recovery; (5) energy-absorbing landing; (6) post-landing rescue. The most extreme mode shall not be required for manageable failures.

2.5 Human Authority

Human emergency authority is preserved and non-bypassable. No AI system, optimization algorithm, or predictive model shall override the crew's ability to command, inhibit, or terminate an automatic survival sequence. This is a design requirement, not a guideline.

3. What Is Defined Today vs. What Remains Open

■ This section is the first thing a technical due-diligence team should read.

Domain	What Is Defined	What Remains Open	Gate
System architecture	Six-layer survival; Safety Kernel; state machine; staged separation; 3+2 recovery; 8-support landing	All engineering analysis	G1
Mass accounting	26,000 kg and 7,000 kg planning models at 30.0 m rotor; component-level allocation	Bottom-up verification; uncertainty and growth	G1
Rotor geometry	30.0 m diameter, 706.86 m ² disk area, 36.78 kg/m ² disk loading (CALC)	Rotor structural design and dynamic clearance (R24-OI-02)	G1–G5
Capsule structure	CFRP roll-cage concept; titanium nodes; load-path description	FEA; crash-load analysis; fatigue	G5
Separation system	S0–S5 timeline; candidate non-rotor components; mandatory protections	S3 mechanism; 6-DOF; ground demonstration	G2–G3
Recovery system	3+2 canopy architecture; sizing rationale; failure-tolerance concept	Canopy sizing; inflation dynamics; min. deployment altitude	G6
Landing supports	8-point concept; energy absorber; asymmetric-failure requirement	FEA; drop tests; terrain interaction	G5
Crashworthy seats	≥150 mm stroke; ≤6,670 N criterion; 8 positions	Seat design; foam/stroking-leg; dynamic tests	G2–G5
Airbag system	16-bag concept; 2 per occupant; placement; 200 kg budget	Packaging; deployment logic; sealed-volume gas (OI-18); sled tests	G2–G5
Safety Kernel	Precondition set defined; verification requirements stated	Software/hardware implementation; formal verification	G1–G2
IP claims	IP-A, IP-B, IP-C, IP-D conceptually defined; prior-art posture stated	Professional prior-art search; claim drafting; filing	External legal
Certification basis	CS-29 referenced	Formal regulatory engagement; means of compliance	External regulatory
Performance	Hover power study (≈3.10 MW ideal / ≈4.13 MW @ FM 0.75); disk-loading benchmark	All performance analysis; cruise; OEI; range closure	G1
Mission profile	1,000 km range target; 8 occupants; FULL fuel planning mass ≈2,600 kg	Fuel model; cruise speed; altitude; reserve policy	G1

4. Aircraft Geometry and Configuration

4.1 Rotor Disk Loading — Benchmark Comparison (30.0 m Configuration)

Aircraft	MTOW	Main Rotor Diameter	Disk Loading
Mi-26 (Russia — heaviest production)	56,000 kg	32.0 m	~55 kg/m ²
CH-53K King Stallion (US heavy-lift)	38,400 kg	24.1 m	~58–60 kg/m ²
CH-47F Chinook (tandem, per disk)	22,680 kg	18.3 m per rotor	~43 kg/m ² per disk
Sikorsky S-92 (medium offshore)	12,020 kg	17.17 m	~52 kg/m ²
AW139 (medium — market neighbor)	6,800 kg	13.8 m	~45 kg/m ²
HSH Gen1 Rev23 (superseded)	26,000 kg	26.0 m	48.97 kg/m ²
HSH Gen1 Rev25 (this baseline)	26,000 kg	30.0 m	36.78 kg/m ²

Design rationale: HSH's Rev 25 disk loading of 36.78 kg/m² is deliberately lower than both the Rev 23 baseline (48.97 kg/m²) and the Mi-26/CH-53K band. This trades a larger, heavier rotor system for: (a) greater hover-power margin — needed given the additional mass of the capsule, separation system, airbags and crashworthy seats; (b) reduced downwash velocity at the eight fixed landing supports; and (c) lower tip-speed requirement, reducing rotor noise. The cost is paid in main-rotor system mass (+800 kg) drawn from the growth/configuration reserve rather than by increasing design mass. To be confirmed or revised at Document 063 closure.

4.2 Aircraft Geometry Summary

Geometry Item	Preliminary Definition	Status
Overall length	Approx. 16.5–18.0 m	TARGET/STUDY
Maximum fuselage width	Approx. 4.0–4.5 m	TARGET/STUDY
Overall height	Approx. 4.8–5.3 m	TARGET/STUDY
Main rotor diameter	30.0 m	TARGET — REV24/25
Main rotor radius	15.0 m	CALC
Main rotor disk area	706.86 m ²	CALC
Disk loading	36.78 kg/m ²	CALC
Tail rotor diameter	4.0 m	TARGET
Tail rotor disk area	12.57 m ²	CALC
Rotor arrangement	Single main rotor + tail rotor	REQ/STUDY
Landing supports	8 fixed, short, mechanical supports under capsule	TARGET
Retractable landing gear	None	LOCKED CONCEPT
Capsule position	Central, integrated within primary airframe	LOCKED CONCEPT
Capsule normal-flight load path	Primary attachment system with controlled emergency release	REQ
Capsule emergency load path	Independent final landing element after verified clearance	REQ

4.3 Rotor Geometry Calculation Detail

$D = 30.0 \text{ m}$ [TARGET]. $R = D/2 = 15.0 \text{ m}$. $A = \pi R^2 = \pi \times 225 = 706.8583 \text{ m}^2$ [CALC]. At $M = 26,000 \text{ kg}$, disk loading $\delta = 26,000 / 706.86 = 36.7825 \text{ kg/m}^2$ [CALC]. Compared with the Rev 23 configuration ($A = 530.93 \text{ m}^2$, $\delta = 48.97 \text{ kg/m}^2$), disk area increases by a factor of $(30/26)^2 \approx 1.331$ and disk loading decreases by approximately 24.9%.

5. Baseline Status and Configuration Control

5.1 Status Label Definitions

Status Label	Precise Meaning
[LOCKED CONCEPT]	Retained as baseline design principle; not to be changed without formal impact assessment
[LOCKED LIMIT]	Hard Generation 1 constraint; requires formal change process and full re-analysis to alter
[REQ]	Top-level requirement; binding on the design unless formally waived
[TARGET]	Numerical or functional development objective; not demonstrated
[ASSUMPTION / STUDY]	Working engineering assumption for preliminary sizing; to be replaced by analysis or test
[CALC]	Calculated from stated assumptions; inputs are themselves assumptions or targets
[OPEN]	Requires analysis, simulation, test, supplier data or design decision
[OPEN — CRITICAL]	Open item with material effect on the core safety case; priority closure required
[DEFERRED — RESEARCH TRACK]	Excluded from licensable core baseline; not to be presented as core product
[TBD — DEPENDENCY]	Cannot be fixed until a dependent activity closes
[PRIOR ART STATUS: NOT YET SEARCHED]	No novelty conclusion implied; professional search required

5.2 Configuration Change Triggers

A formal configuration change is required for any of the following: MTOW; main rotor diameter; tail rotor diameter; capsule mass; occupant count; separation-sequence architecture; parachute architecture; landing-support count or geometry; airbag count or placement; seat-stroke specification; Safety Kernel precondition set; any safety-layer definition. Each change triggers an impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification. Rev 24/25 is itself such a change for main rotor diameter and associated mass lines, and its impact assessment is summarized in §5.3.

5.3 Change Impact Assessment Summary (Rev 24/25 Rotor Change)

Affected analyses: rotor structural loads, hub design, transmission torque path, ground resonance, blade sailing, dynamic clearance to airframe and ground, hover/OEI margins, noise, and mass/CG. Affected documents: any tree document citing 26.0 m, 530.93 m², 48.97 kg/m², or the 2,100 kg rotor mass line, or the Rev 23 hover STUDY figures, must be aligned to the Rev 25 30.0 m configuration before further generation.

PART II

Safety and Survival Architecture

Six-layer survival chain, Safety Kernel, separation system, capsule, landing supports, airbags, seats, recovery canopies, core safety case, timeline, state machine, failure modes, interfaces

6. Six-Layer Survival and Safety Architecture

The six layers are functionally distinct and independently capable — not merely redundant. Independence means physically, electrically, and software-separate implementations that cannot both be defeated by any single common-cause failure. Duplication of components does not by itself establish independence; architectural, physical, electrical, software, power, sensor, interface, and common-cause analyses are required before claiming credit for any layer.

Layer	Function	Primary Purpose	Independence Requirement
L1	Primary automatic emergency detection and initiation	Detect catastrophic states; initiate validated survival sequence when Safety Kernel authorizes	Separate processor, sensor suite, power bus from L2
L2	Independent backup initiation	Second segregated automatic path if L1 unavailable or in disagreement	Different physical location, software architecture, power source from L1
L3	Independent manual/crew authority	Crew initiation, inhibit and emergency authority without L1/L2	Hardwired path bypassing L1/L2; software cannot disable
L4	Rapid selective physical separation	Release designated non-rotor hazards using controlled mechanisms	Dual independent actuation paths; no single-point failure defeats both
L5	Direct-impact survival	Protect occupants via capsule structure, restraints, seats and eight energy-absorbing supports	Passive structural system; no active components required for primary function
L6	Recovery and post-event survival	Deploy recovery canopies; locate occupants; support rescue	Independent emergency power; independent comms; independent locator beacon

■ AI Boundary — Non-Negotiable Principle: AI or advanced prediction may assist detection, classification and decision support. Deterministic safety logic authorizes safety-critical actuation. AI confidence is never equivalent to safety authorization. This principle is non-negotiable from a certification perspective and is the engineering basis for IP-A (Part III, Section 17).

Survival Chain

PREVENT/MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE (NON-ROTOR) → VERIFY CAPSULE CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT (8 supports + crashworthy seats + 16 airbags) → SURVIVE / LOCATE / RESCUE → AUDIT

7. Deterministic Safety Authorization and Safety Kernel

The Safety Kernel prevents AI, digital-twin models, and predictive systems from directly commanding hazardous separation without satisfying a defined set of physically verifiable preconditions. It is the boundary between the decision layer and the actuation layer.

7.2 Safety Kernel Precondition Set

Precondition Category	Required Evaluation
Altitude and airspeed	Above minimum recovery altitude; within separation dynamic-pressure envelope; within parachute deployment envelope accounting for sequence time
Attitude and angular rates	Pitch, roll, yaw within acceptable separation envelope; angular rates within acceptable separation rate limits
Rotor and engine state	Rotor RPM documented; engine/fire/fuel hazard state known and consistent with separation sequence
Structural and attachment	Capsule attachment health confirms release mechanisms functional; structural health consistent with planned separation
Trajectory and clearance	Acceptable clearance between released components and capsule/parachutes confirmed (real-time or pre-computed envelope), including clearance relative to the 30.0 m rotor disk
Recovery system	Parachute deployment envelope satisfiable after separation; recovery integrity confirmed
Power and communication	Independent survival power available; communication loss does not inhibit Safety Kernel
Human authority	Manual crew inhibit confirmed; rollback/fallback availability assessed
Airbag authorization	Impact state valid; deployment permitted under deterministic preconditions; capsule-interior environment suitable

7.3 Kernel Decision Table

Decision	Minimum Design Question	Verification Method
Authorize separation	Are all mandatory preconditions simultaneously true?	Formal logic review + fault injection
Inhibit separation	Can an unsafe state block actuation even under software fault?	Fault injection + power-loss test
Fallback mode	What happens if a sensor or command path is lost?	Failure-mode testing per FMEA
Human override	Is authority bounded, deterministic and physically independent?	HMI + system integration test
Airbag authorize	Is impact state valid and deployment safe in closed capsule?	Logic review + sled/impact test + gas analysis
Audit record	Can every emergency event be fully reconstructed?	Data-recording verification test

The Safety Kernel shall be implemented with: formally verified software or hardware logic; documented requirement set with code traceability; independence from AI/ML layers; segregated power supply; independent communication path to actuation hardware; protection against command injection; and an immutable audit record. No update or in-flight reprogramming shall modify the Safety Kernel precondition set without a formal change process and re-verification.

8. Controlled Rapid Separation System

The system is selective, staged, and controlled — not a blanket jettison of everything. State-dependent release logic means the set of components released in a fire emergency differs from the set in a structural emergency. Every component designated as separable must be confirmed through structural, aerodynamic, dynamic, fire, control, trajectory and collision analysis.

8.2 Candidate Separable Components

Component	Condition for Separation	Analysis Requirement
Main engines (both)	After fuel isolation; where structurally separable	Trajectory, structural, thermal, fire analysis
Selected transmission/drive components	Where release is structurally and dynamically acceptable	6-DOF dynamics, CG-shift analysis
Tail rotor assembly + gearbox + drive	Where aerodynamically and dynamically safe	Anti-torque state, trajectory, collision analysis
Primary fuel tanks + major fuel lines	After confirmed fuel isolation	Fire-hazard analysis, isolation timing
Selected external structural panels	High-mass, not on capsule primary load path	Structural, trajectory, collision analysis
Main rotor blades and hub	EXCLUDED from core candidate list	Deferred research track — §12

8.3 Mandatory Release Protections

Protection	Requirement
Positive mechanical retention	Active positive lock during all normal flight; release requires deliberate actuation
Electrical/logic inhibit	Release electrically inhibited in normal flight state; specific arming sequence required
Independent command paths	Primary and backup paths physically and electrically independent
Defined arm/fire sequence	Documented, tested, protected against out-of-sequence commands
Release-state sensing	Positive confirmation of attached/released state at each stage
Fuel/energy isolation	Isolation precedes or is concurrent with release; no new uncontrolled hazard created
Asymmetric/partial-release detection	Full reporting of any case where less than the designated set is released
Post-release monitoring	Track released components where practically possible
Capsule-clear verification	Capsule release not authorized until all non-rotor components confirmed clear
Failed-release fallback	Safe state defined for every possible failed-release configuration

8.4 Separation System Mass Budget — 1,000 kg (Unchanged)

Separation System Component	Mass
Primary capsule release mechanisms and locks	240 kg
Independent backup release mechanisms	150 kg
Local structural reinforcement and release fittings	160 kg
Independent initiators / actuators	90 kg
Release-state sensors and monitoring	65 kg
Independent power / signalling and protected wiring	70 kg
Fuel / energy isolation interfaces	55 kg
Retention, inhibit and safety-interlock hardware	50 kg
Configuration growth allowance + integration margin	120 kg
TOTAL	1,000 kg

9. Occupied Survival Capsule

The capsule is not a cabin with a parachute. It is a structurally independent survival system that functions as the passenger compartment during normal flight and becomes the final landing boundary in a catastrophic emergency. Every subsystem inside the capsule must function independently of the main airframe after separation. Primary structure: continuous CFRP roll-cage with metallic/titanium nodes at load-introduction points. CFRP provides high specific strength; titanium nodes provide fatigue-resistant, corrosion-resistant, impact-resistant interfaces at maximum stress-concentration points. Continuous roll-cage geometry provides controlled deformation paths — progressive outward deformation rather than inward collapse. The capsule architecture and mass target are unchanged by the Rev 24/25 rotor-diameter configuration change.

Capsule Parameter	Rev 25 Definition	Status
Occupancy	8 persons maximum	LOCKED LIMIT
Complete capsule mass	7,000 kg planning target	TARGET
Primary structure	Continuous CFRP roll-cage, titanium nodes	CONCEPT
Seat type	Energy-absorbing, ≥ 150 mm stroke, $\leq 6,670$ N peak spinal load	TARGET/STUDY
Airbag system	16 total (2 per occupant)	TARGET/STUDY
Landing supports	8 fixed, progressive energy absorbers	TARGET
Recovery canopy architecture	3 primary + 2 reserve	STUDY
Independent emergency power	Battery-based, independent of airframe bus	TARGET
Life support	Emergency oxygen, minimum post-landing endurance	TARGET
Capsule-clear verification	Required before recovery deployment	REQ
Post-landing stability	8-point architecture; asymmetric/failed-leg analysis required	REQ
Water landing	Flotation, sealing for ditching scenario	TARGET
Emergency egress	Multiple exits rated for crash loads; airbag-compatible paths	TARGET

10. Eight Fixed Short Mechanical Landing Supports

Eight short, thick, fixed mechanical supports are integrated with the capsule lower structure. They are not conventional helicopter landing gear. Key rationale: capsule-integrated — supports remain with occupants after separation; redundancy — loss of one or two supports still leaves six or seven functional; energy absorption — each support incorporates a progressive mechanical energy-absorber with controlled, bounded crush force throughout deformation without a hard metallic stop.

Impact Velocity	Total KE (7,000 kg capsule)	KE per Support ($\div 8$)
6 m/s	126.0 kJ	15.75 kJ
8 m/s	224.0 kJ	28.0 kJ
10 m/s	350.0 kJ	43.75 kJ

■ These are sizing illustrations, not allowable impact energies. Actual survival depends on pulse shape, attitude, angular rates, terrain and structural response. Seat stroke and airbags provide additional occupant load management after the supports arrest capsule motion.

Case	Analysis Focus	Required Evidence
All 8 supports, flat terrain	Load sharing and stability	FEA + drop test
1 support failed	Redistribution and local overload	FEA + test
2–3 supports failed	Survivability boundary	Sensitivity study
Sloped terrain ($\leq 15^\circ$)	Roll/tip tendency, asymmetric load transfer	Terrain simulation
Soft ground / snow	Penetration and energy dissipation	Ground interaction test
Water impact (ditching)	Hull sealing and flotation	Water impact test
Oblique combined loads	Lateral + vertical + longitudinal	6-DOF dynamic simulation

10A. Sixteen-Airbag Occupant Protection Architecture

Item	Rev 25 Baseline
Occupant positions	8 (locked maximum)
Airbags per occupant	2
Total airbags	16
Primary set	8 — one per seat, head/torso / principal impact direction
Secondary set	8 — one per seat, lateral / lower-body / secondary direction
Status	TARGET/STUDY
Mass planning (corrected)	≈ 200 kg (16 aviation-grade modules + inflators + ECU + pyrotechnic harness + mounting)
Verification	Sled tests, inflation tests, interference checks, gas-toxicity review, egress compatibility

Placement principles. Eight primary + eight secondary; shall not obstruct egress; compatible with CFRP roll-cage, seats and restraints; deployment only under deterministic Safety Kernel preconditions; airbag authorization passes through the Safety Kernel — not an independent system that can deploy outside the authorization chain.

Mass correction rationale. Rev 19 estimated 80–120 kg. Aviation-grade modules at 1.5–2.0 kg each $\times 16 \approx 24$ –32 kg module mass. Adding ECU (8–15 kg), pyrotechnic harness (20–30 kg), mounting (30–40 kg), wiring (15–20 kg), and qualification margin yields approximately 150–220 kg, rounded to 200 kg. The Rev 19 estimate was insufficient and would have created a false closure of the capsule mass budget.

■ Sealed-Capsule Deployment Hazard — OI-18 (CRITICAL). Sixteen airbags deploying simultaneously in a sealed capsule of limited internal volume create risks not present in automotive applications: (a) overpressure — rapid gas generation may transiently exceed safe ear-drum and sinus pressure limits; (b) gas composition — propellant gases dilute capsule oxygen; (c) temperature — inflator heat in a sealed volume; (d) particulate — sodium compounds from certain propellant chemistries. Mitigation options include staggered sequential deployment, vented inflators, or reduced inflator energy for the closed-capsule environment. Must be analyzed before deployment logic is locked.

10B. Eight Crashworthy Seats — Corrected Specification

Rev 19 specified “ ≥ 50 cm ergonomic foam seating-surface thickness.” This is not a recognized aviation crashworthy-seat criterion. FAR 27.562 (and the CS-29 equivalent) evaluates seats by occupant injury criteria, not foam thickness. Soft foam can compress fully under crash loads in milliseconds, providing minimal protection after initial compliance. The correct engineering specification is a minimum dynamic stroke and a peak load limit.

Item	Rev 25 Baseline
Number of seats	8
Minimum dynamic stroke	≥ 150 mm (combined seat/absorber/floor deformation path)
Peak spinal compressive load limit	≤ 6,670 N (FAR 27.562 reference criterion, proxy)
Force-displacement response	Progressive — bounded crush force throughout stroke, no hard stop at stroke limit
Seat architecture	Energy-absorbing leg + foam/elastomer cushion combination [OPEN — detailed design]
Function	Limit peak loads transmitted to pelvis, spine and lower extremities during hard landing; works with 8 landing supports + 16 airbags + 5-point restraints + roll-cage
Status	TARGET/STUDY — density curve, stroke profile, fire rating, mass, dynamic performance [OPEN]
Mass	Included in 700 kg seats + restraints capsule line
Verification	Dynamic seat tests per FAR 27.562 or CS-29 equivalent; combined drop/sled cases

11. Recovery Architecture — 3 Primary + 2 Reserve Canopies

Three large primary canopies are preceded by crewed spacecraft recovery systems at comparable mass. Fewer, larger canopies reduce simultaneous-inflation complexity and inter-canopy collision risk at low altitude — the failure mode most common in real rotorcraft emergencies.

Recovery Mode	Use Case	Status
Controlled flight recovery	Recoverable disturbance	REQ/STUDY
Autorotation	Engine/power loss when altitude/rotor state permit	REQ/STUDY
Selective mass/hazard separation	Fire, structural or combined emergency	REQ/CONCEPT
Capsule separation	Catastrophic state	REQ/CONCEPT
3 primary recovery canopies	Primary post-separation vertical recovery	STUDY
2 reserve canopies	On detected primary failure only	STUDY
8 energy-absorbing landing supports	Final ground-impact attenuation	REQ/TARGET
Crashworthy seats + 16 airbags	Occupant load management after ground contact	TARGET/STUDY
Post-landing survival	Locator, comms, lighting, emergency power, rescue interface	REQ

For a 7,000 kg capsule at a 7.5 m/s target descent rate: total canopy area $A_{\text{total}} \approx 2,983 \text{ m}^2$; per canopy ($\div 3$) $\approx 994 \text{ m}^2$ each; corresponding diameter $\approx 35\text{--}36 \text{ m}$ per primary canopy. Consistent with spacecraft precedent, including Soyuz-class multi-main canopy recovery. Detailed sizing is deferred to Document 159. This sizing illustration is unaffected by the main-rotor diameter change since it depends only on capsule mass and target descent rate.

■ The minimum deployment altitude is [OPEN — CRITICAL] (OI-02). Most rotorcraft accidents occur at low altitude.

12. Core Safety Case — Rebuilt Without Main-Rotor Separation

The single largest credibility risk in Rev 12 was the implicit dependency of the safety case on main-rotor separation — a capability for which no credible 6-DOF simulation, structural trajectory analysis, or physical evidence exists. A 30.0 m rotor carries substantially more rotational kinetic energy and a larger gyroscopic moment resisting rapid attitude change than the Rev 23 26.0 m rotor. Removing this dependency from the core safety case is what makes the remaining architecture defensible, and the Rev 25 rotor enlargement makes this exclusion more, not less, important.

The survival chain works assuming the main rotor is never separated.

- 1 Controlled flight recovery / autorotation — no separation required.
- 2 Selective non-rotor hazard separation — engines, fuel tanks (post-isolation), tail rotor/drive, fire-load panels.
- 3 Capsule release from airframe — capsule departs beneath/behind the still-attached rotor disk.
- 4 Capsule descent under the 3+2 canopy system.
- 5 Impact attenuation via eight fixed supports + crashworthy seats + 16 airbags.
- 6 Post-landing survival — independent power, life support, locator, communications.

Main-rotor separation is retained as Document 080 deferred research, explicitly excluded from the licensable core baseline and from primary IP claims. If independent 6-DOF analysis and mechanism-test evidence later demonstrate safe separation, it may be reintroduced through the configuration-change process.

13. Staged Separation Timeline

Stage	Action	Target Window	Confidence Basis	Status
S0	Validate emergency state; arm path; isolate fuel/electrical/energy	0.0–0.5 s	Comparable to ejection-seat arming	TARGET
S1	Fuel isolation and designated fire-hazard separation	0.5–1.8 s	Comparable to crash fuel-shutoff	TARGET
S2	Engine + tail-rotor drive separation (non-main-rotor)	1.8–3.5 s	Comparable to cargo-release, scaled	TARGET/OPEN
S3	Capsule-to-airframe primary structure release	3.5–5.0 s	No direct certified analog	OPEN — CRITICAL
S4	Capsule clearance verification	5.0–5.8 s	Sensor-response-limited	TARGET
S5	Recovery canopy authorization	After S4, gated on clearance	Not time-gated; correctness first	REQ

■ If the architecture cannot safely achieve these windows, targets shall be revised under configuration control — not achieved by reducing verification rigor. Partial and asymmetric separation are first-class emergency states at every stage. Clearance geometry relative to the larger 30.0 m rotor disk after release may affect S4 trajectory analysis and is flagged under R24-OI-02.

14. Emergency State Machine

State	Entry Condition	Primary Action	Exit / Next
1 — NORMAL FLIGHT	All systems nominal	Continuous monitoring; locks engaged	Anomaly → 2
2 — ANOMALY DETECTED	Sensor/model flags abnormality	Increase monitoring; begin confirmation	Confirmed → 3; Rejected → 1
3 — CONFIRMATION	Potential condition detected	Independent evidence and validity checks	Confirmed → 4
4 — IMPACT ASSESSMENT	Condition confirmed	Estimate authority, altitude, airspeed, fire/structural risk	Complete → 5
5 — RECOVERY ATTEMPT	Assessment complete	Conventional control, OEI, autorotation	Succeeds → 1; Fails → 6
6 — SURVIVAL AUTHORIZATION	Recovery failed	Safety Kernel evaluates all preconditions	Authorized → 7; Not met → degraded

State	Entry Condition	Primary Action	Exit / Next
7 — STAGED SEPARATION	Authorized	Execute designated release set (non-rotor)	Complete → 8
8 — CAPSULE CLEARANCE	Separation executed	Verify clearance; identify residual attachments	Clear → 9; Partial → partial-release
9 — RECOVERY DEPLOYMENT	Clearance verified	Authorize parachute within validated envelope	Deployment confirmed → 10
10 — DESCENT MANAGEMENT	Canopies deployed	Monitor attitude, rates, canopy state, support readiness	Ground imminent → 11
11 — LANDING ATTENUATION	Ground contact	Supports + seats + airbags all load simultaneously	Landing complete → 12
12 — POST-LANDING SURVIVAL	Stabilized	Locator active; comms active; life support active	Rescue confirmed → 13
13 — AUDIT	Rescue complete	Preserve all event data, sensor states, command history	Data secured — END

15. Independence, Failure Modes and Common-Cause Control

Failure / Hazard	Severity	Required Design Response
Unintended capsule separation in normal flight	Catastrophic	Mechanical locks + electrical inhibits + command validation + monitoring + quantitative safety objective
Partial separation (incomplete component release)	Critical	State sensing + asymmetric-load analysis + defined degraded recovery mode
Asymmetric separation	Critical	6-DOF dynamic analysis + CG/inertia update + post-release control assessment
Release mechanism failure	Critical	Independent path + defined safe fallback; no assumption of perfect release
Sensor loss or bad data	Hazardous	Validity monitoring + redundant/diverse sensing + safe-state logic
L1 automatic detection failure	Hazardous	Independent L2 backup + L3 manual crew initiation
Power-bus failure	Critical	Independent survival power for critical functions
Communication loss	Hazardous	Local deterministic safety logic; communication not in authorization path
AI/ML model failure	Hazardous	Deterministic Safety Kernel remains authoritative
Fire during separation	Critical	Fuel isolation + fire containment + coordinated sequencing
Released component collision with capsule/parachutes	Catastrophic	Trajectory analysis + clearance envelopes + controlled sequencing
Parachute deployment outside envelope	Critical	Hard inhibits + validated deployment conditions
Landing-support failure (one or more)	Hazardous	8-support redundancy + asymmetric/failed-leg structural analysis
Airbag inadvertent deployment in flight	Critical	Deterministic inhibit logic + dual-path sensing + mechanical safing
Airbag non-deployment on impact	Hazardous	Backup initiation path + sled test verification
Airbag overpressure / gas toxicity in sealed capsule	Critical	OI-18: sealed-volume gas analysis required

Failure / Hazard	Severity	Required Design Response
Seat stroke bottom-out at extreme load	Hazardous	Progressive foam/absorber + support stroke + analysis against worst-case velocity
Common-cause EMI/lightning	Critical	CCA + physical/electrical segregation + zone separation
Common-cause fire	Critical	Fire zone segregation + barriers between survival-critical systems
Larger-rotor dynamic clearance after release (R24-OI-02)	Hazardous	6-DOF clearance analysis at 30.0 m disk geometry

16. Principal System Interfaces

ID	Definition	Normal-State	Emergency Function	ICD
CAP-AF	Capsule ↔ airframe structural attachment	Transmit normal flight loads	Controlled release on Safety Kernel authorization	Doc 136
CAP-PAR	Capsule ↔ parachute system	Parachutes stowed; deployment inhibited	Deploy primary canopies after clearance verified	Doc 164
CAP-LG	Capsule ↔ eight landing supports	Ground loads during normal operations	Primary crash/landing load path after separation	Doc 156
CAP-AB	Capsule ↔ 16 airbags + sensors/inflators	Stowed; armed only under deterministic preconditions	Deploy under Safety Kernel authorization at impact	Doc 145
CAP-SEA T	Capsule floor ↔ 8 crashworthy seats + restraints	Normal seating load path	Stroke + energy absorption during impact	Doc 123
JET-PWR	Independent release power bus	Standby / trickle charge	Power to all release mechanisms on command	Doc 407
JET-CMD	Automatic/backup/manual command paths	Inhibited; monitoring only	Transmit release commands through independent paths	Doc 406
JET-SNS	Release-state sensing	Report attached state	Confirm released state at each stage	Doc 408
FUEL-IS O	Fuel isolation ↔ separation coordination	Normal fuel flow	Isolate fuel before or concurrent with separation	Doc 104
REC-LO C	Post-landing locator/communications	Dormant / periodic beacon check	Activate post-separation; transmit position	Doc 405
AV-CTRL	Avionics/control ↔ Safety Kernel	Pass flight data for monitoring	Safety Kernel issues authorization; avionics executes	Doc 241

Intellectual Property and Competitive Position

Candidate IP claims, prior-art posture, IP-to-engineering traceability, and competitive differentiation

17. Engineering / IP Value Surfaces

Rev 20/22/23/25 concentrates on four primary claim candidates and retains remaining elements as supporting system narrative. Four narrowly-scoped, well-supported claims are stronger than nine broad assertions. None of the four claims is rotor-diameter-dependent, so the Rev 24/25 configuration change does not alter the IP posture described below.

IP-A — Deterministic Safety Kernel Gating AI-Assisted Emergency Prediction

The specific combination of: (a) AI/ML-based anomaly prediction; (b) a hard deterministic precondition layer; and (c) selective, state-dependent (not all-or-nothing) actuation in a crewed rotorcraft survival system. Assessed as the strongest single claim in the package.

IP-B — Eight-Point Fixed Capsule-Integrated Landing Support with Asymmetric Failure State Management

The combination of: (a) eight-point, capsule-integrated, fixed, energy-absorbing mechanical support configuration; (b) individual support failure-state monitoring; and (c) state management adapting stability assessment to asymmetric failure configurations.

IP-C — State-Dependent Non-Rotor Selective Separation Sequencing for an Occupied Survival Capsule

Staged, state-dependent sequencing logic for selective release of non-rotor hazards from an occupied survival capsule, with explicit exclusion of the main rotor assembly and capsule-centric clearance verification gating recovery deployment. Distinguished from Ka-50/Ka-52 prior art.

IP-D — Integrated Multi-Layer Survival Chain for Crewed Rotorcraft

The specific combination of: (a) Deterministic Safety Kernel (IP-A); (b) state-dependent non-rotor selective separation (IP-C); (c) eight-point capsule-integrated impact attenuation with asymmetric failure management (IP-B); and (d) distributed canopy recovery — as a unified, interdependent system architecture applied to a crewed rotorcraft survival capsule. System-level claims are harder to design around. Prior art (Ka-50: rotor separation + ejection seat; Soyuz: capsule + canopies) cannot anticipate the specific four-element combination. Assessed as the most commercially valuable claim in the package.

■ Supporting narrative (not standalone patent candidates): six-layer taxonomy; multi-mode recovery; partial/asymmetric-release management; immutable emergency event record; 16-airbag layout; crashworthy seat stroke specification. No novelty, patentability or freedom-to-operate conclusion is asserted by this baseline.

18. Risk Register

Risk	Severity	Current Status	Priority
Stage S3 (capsule-to-airframe release) has no direct certified engineering analog	High	Open research item	P0 Critical
Low-altitude canopy deployment envelope may exclude most real accident profiles	High	Open — closure study required	P0 Critical
7,000 kg capsule mass not yet bottom-up verified	Medium	Planning allocation only	P0 Critical
No 6-DOF simulation evidence for any separation stage	High	Gate G4 not started	P0 Critical
Capsule clearance from 30.0 m rotor disk after separation not analyzed	High	Not yet addressed — R24-OI-02	P0 Critical
16-airbag deployment in sealed capsule — overpressure and gas-toxicity not analyzed	Critical	OI-18	P0 Critical
Airbag system mass 200 kg planning — detailed design may exceed budget	Medium	Corrected estimate	P0 High
Crashworthy seat design — stroke, load-distribution, combined-load cases not closed	Medium	TARGET/STUDY	P0 High
Six-layer independence against common-cause not demonstrated	High	CCA not performed	P0 Critical
30.0 m rotor structural design, blade/hub BoM, dynamic clearance and ground resonance unverified	High	R24-OI-01 / R24-OI-02	P0 Critical
Growth/configuration reserve reduced to 1,200 kg after funding rotor mass uplift	Medium	Programme reserve consumed	P1 High
No professional prior-art search for IP-A, IP-B, IP-C, IP-D	Medium	Prior-art posture only	P1 High
Certification basis not formally engaged with any regulator	Medium	Conceptual reference	P1 High
Main-rotor separation (deferred) remains structurally unanalyzed	Medium	Deferred research track	P2 Medium

19. Prior Art Posture

Class 1 — Ejection-seat rotorcraft (Ka-50/Ka-52)

Rotor-blade-severing + crew ejection is directly relevant prior art for any main-rotor-related separation claim. This is why main-rotor separation is excluded from all primary IP claims (IP-A through IP-D).

Class 2 — Crewed spacecraft return capsules

Multi-canopy recovery (Soyuz 3 mains, Crew Dragon 3 mains), crashworthy seating, and independent post-landing life support are prior art for capsule-recovery and life-support elements. IP claims are narrowed to the rotorcraft-specific combination, not the general capsule concept.

Class 3 — Helicopter energy-absorbing landing gear

Energy-absorbing skids and struts exist in certified rotorcraft. IP-B novelty rests on the eight-point capsule-integrated configuration and individual support failure-state management — not on energy absorption in general.

Class 4 — Autonomous vehicle safety interlocks (IEC 61508, ISO 26262, DO-178C)

Deterministic safety interlocks are well developed. IP-A novelty is not in the concept of a safety interlock but in the specific combination of AI prediction + deterministic kernel + selective capsule-centric actuation in a crewed rotorcraft

survival system.

Class 5 — Automotive and aviation airbag systems

Airbag systems are prior art. Any IP related to airbags would rest on the capsule-specific integration with the Safety Kernel, the sealed-volume environment, and the combined-attenuation architecture.

Class 6 — IP-D system-level combination

No single prior-art reference combines all four elements of IP-D (Safety Kernel + selective non-rotor separation + eight-point capsule attenuation + distributed canopy recovery) simultaneously in a crewed rotorcraft architecture. This is the basis for IP-D's strength. A professional prior-art search against IP-A through IP-D is required before any patent filing or broad non-NDA disclosure.

23. Competitive Differentiation

■ Descriptive comparison, not a claim of superior certified safety performance. All HSH advantages are architectural concepts pending engineering verification.

Feature	Bell 412	AW139	S-92	Mi-26	HSH Rev25
Occupied survival capsule as final landing boundary	X	X	X	X	✓ CONCEPT
Deterministic Safety Kernel (AI-gated)	X	X	X	X	✓ CONCEPT
State-dependent selective non-rotor separation	X	X	X	X	✓ CONCEPT
Eight-point capsule-integrated impact attenuation	X	X	X	X	✓ CONCEPT
Distributed canopy recovery (3+2)	X	X	X	X	✓ STUDY
16-airbag occupant protection in capsule	X	X	X	X	✓ TARGET/STUDY
Six independently-capable survival layers	X	X	X	X	✓ CONCEPT
Independent post-separation life support	X	X	X	X	✓ TARGET
Explicit asymmetric/partial-failure state management	X	X	X	X	✓ CONCEPT
Crashworthy seats (≥150 mm stroke)	Some	Some	Some	Military	✓ TARGET/STUDY
Reduced disk loading (36.78 kg/m ²) at 26,000 kg class	N/A	N/A	N/A	N/A	✓ CALC — Rev 25

Commercial positioning. HSH is not positioned as a replacement for existing helicopter models on cabin-space or payload-fraction grounds. It is positioned as a distinct product category — a survivability-first rotorcraft — for operators whose primary requirement is occupant survival in catastrophic emergencies: offshore energy, search-and-rescue, medical evacuation, governmental/military VIP transport.

PART IV

Critical Open Items Register

The complete OI-01 through OI-18 register plus the Rev 24/25 rotor-configuration open items, closure paths, and the IP-to-engineering traceability matrix

20. Critical Open Items Register

ID	Open Item	Status	Pri	Closure Path
OI-01	Stage S3 capsule-to-airframe release has no certified engineering analog	OPEN — CRITICAL	P0	Mechanism design + 6-DOF + ground demo
OI-02	Minimum reliable canopy deployment altitude for 7,000 kg capsule not closed	OPEN — CRITICAL	P0	Sizing + inflation dynamics + envelope study
OI-03	7,000 kg capsule mass remains a planning allocation	OPEN — CRITICAL	P0	Component mass database + weighing plan
OI-04	No 6-DOF simulation for any separation stage	OPEN — CRITICAL	P0	Model hierarchy + correlation plan (G4)
OI-05	Partial and asymmetric separation detection and degraded recovery not detailed	OPEN — CRITICAL	P0	State sensing + FMEA + test cases
OI-06	Independence of six safety layers against common-cause not demonstrated	OPEN — CRITICAL	P0	CCA + fault injection + power/sensor diversity
OI-07	Capsule clearance from rotor disk after separation not analyzed (30.0 m disk)	OPEN — CRITICAL	P0	6-DOF clearance analysis
OI-08	Occupant injury criteria and seat/restraint dynamic performance not closed	OPEN	P0	Dynamic seat tests + criteria selection
OI-09	Eight-support asymmetric failure and terrain interaction cases not closed	OPEN	P0	FEA + drop test matrix
OI-10	26,000 kg vehicle mass lacks component-level uncertainty tracking	OPEN	P1	Mass control plan + reserve policy
OI-11	Rotor/propulsion power-available vs power-required (OEI) not closed	OPEN	P1	Performance model + engine data
OI-12	Fuel isolation timing coordinated with separation not closed	OPEN	P1	Hazard analysis + sequence timing
OI-13	Professional prior-art search for IP-A through IP-D not performed	OPEN	P1	External IP counsel
OI-14	Certification basis not formally engaged	OPEN	P1	Regulator engagement plan
OI-15	ICDs for CAP-AF, CAP-PAR, CAP-LG, CAP-AB, CAP-SEAT, JET-* not issued	OPEN	P1	ICD set under configuration control
OI-16	16-airbag packaging, placement, mass, inflators, deployment logic not closed	OPEN	P0	Design + sled/impact tests
OI-17	Crashworthy seat stroke and load design (≥ 150 mm, $\leq 6,670$ N) not closed	OPEN	P0	Seat design + dynamic tests

ID	Open Item	Status	Pri	Closure Path
OI-18	16-airbag deployment in sealed capsule — overpressure and gas-toxicity not analyzed	OPEN — CRITICAL	P0	Sealed-volume gas analysis + toxicity + venting/staging
R24-OI-01	Bottom-up main-rotor BoM vs 2,900 kg planning line	OPEN	P0	Rotor BoM + design definition
R24-OI-02	30.0 m rotor dynamics, dynamic clearance, ground resonance	OPEN	P0	Rotor structural + dynamics analysis
R24-OI-03	Cruise energy model for 1,000 km range with FULL fuel	OPEN	P1	SFC + cruise power + reserve model
R24-OI-04	Align HSH-001/002/003 and lower-tier documents to Rev 25 rotor/mass figures	TBD — DEPENDENCY	P1	Document tree reconciliation
R24-OI-05	Installed engine power class vs reduced hover demand at 30.0 m rotor	STUDY/OPEN	P1	Engine selection trade study

33. IP-to-Engineering Traceability Matrix

This matrix connects the four principal IP surfaces to the engineering domains that must close before any claim can be treated as more than conceptual. It is unaffected by the Rev 24/25 rotor-diameter change.

IP ID	Claim Surface	Engineering Domains That Must Close	Primary Tree Anchors	Critical Dependencies
IP-A	Deterministic Safety Kernel gating AI-assisted emergency prediction	Safety Kernel requirements; precondition set; independence from AI/ML; fault injection; software/hardware assurance	SAF / AV / VER series	OI independence + formal verification
IP-B	Eight-point fixed capsule-integrated landing support with asymmetric failure state management	Support geometry; energy absorbers; failed-leg FEA; drop tests; terrain cases	Landing-support / crash / STRUCT	OI-09 + G5 evidence
IP-C	State-dependent non-rotor selective separation sequencing for occupied survival capsule	Candidate component list; S0–S5; release mechanisms; clearance verification; 6-DOF	SEP / DYN series	OI-01, OI-04, OI-05, OI-07
IP-D	Integrated multi-layer survival chain (Kernel + selective separation + eight-point attenuation + distributed canopy recovery)	All of IP-A/B/C evidence plus recovery envelope and combined system demonstration	Cross-domain G3–G6	System-level integration gates

34. Buyer Due-Diligence Readiness Matrix

Intended to let an industrial acquirer review the baseline without reconstructing the entire programme architecture from the full document tree.

Due-Diligence Question	HSH-00 Answer	Primary Supporting Documents	Closure State
What is the core system proposition?	Six-layer survival architecture centered on an occupied survival capsule, deterministic Safety Kernel, selective non-rotor separation, 8-point impact attenuation and 3+2 recovery, at a 30.0 m rotor configuration.	HSH-00 Part I §2, Part II §6, §12; Part VIII §30	Defined
What is technically defined today?	Top-level geometry (30.0 m rotor) and mass allocations, occupancy lock, Safety Kernel preconditions, S0–S5 concept, airbag/seat architecture, IP-A–D posture, OI-01–OI-18, R24-OI-01–05	HSH-00 Parts I–IX	Defined / Studied
What is not yet proven?	The open-item register remains the evidence gap, especially S3 release, low-altitude recovery, mass bottom-up, 6-DOF, OI-18 gas analysis, and 30.0 m rotor structural/dynamic evidence	HSH-00 Part III §18, Part IV §20	Open
What is the evidence path?	G0 architecture freeze through G8 certification/qualification; E0–E5 evidence ladder	HSH-00 Part VII §26, §37; VER series	Defined / Not started
What are the principal residual risks?	S3 release, low-altitude recovery envelope, 7,000 kg mass closure, sealed-capsule airbag environment, six-layer independence, 30.0 m rotor dynamics, reduced 1,200 kg growth reserve	HSH-00 Part III §18, Part IV §20	Open / Critical
What is the IP posture?	IP-A through IP-D are candidate conceptual surfaces with prior-art awareness; not filed; professional search required	HSH-00 Part III §17, §19	Conceptual / Legal open
What configuration is locked?	Generation 1 maximum occupancy of 8 persons; no main-rotor separation in core safety case; main rotor diameter locked at 30.0 m	HSH-00 Part I §1, §2; Part VIII §28	Controlled

Performance, Technology Readiness and Mass Accounting

Hover-power and mission analysis at the 30.0 m rotor configuration, TRL assessment, mass/CG breakdown, and preliminary capsule geometry

21. Preliminary Performance Analysis

21.1 Hover Power Study Calculation

Using actuator disk theory: $P_{ideal} = T^{3/2} / \sqrt{2\rho A}$, where $T = 26,000 \times 9.81 = 255,060 \text{ N}$; $\rho = 1.225 \text{ kg/m}^3$ (ISA sea level); $A = 706.86 \text{ m}^2$ at the Rev 25 30.0 m rotor diameter. Induced velocity $v_i = \sqrt{T/(2\rho A)} \approx 12.136 \text{ m/s}$. $P_{ideal} \approx 3.095 \text{ MW}$ [CALC] — a study value; figure of merit $FM = 0.75$ gives $P_{shaft} \approx 4.127 \text{ MW}$ [CALC]. Installed engine power must exceed P_{shaft} to account for profile power, transmission losses (~4–8%), tail-rotor power (~8–12% of main rotor), accessory extraction, OEI margins, and hot/high performance. A plausible total installed power requirement remains in the range of 5.5–7.5 MW, corresponding to two turboshaft engines of approximately 2,750–3,750 kW each — a reduced power class relative to the Rev 23 26.0 m configuration.

Step	Calculation	Result
1	$T = 26,000 \times 9.81$	255,060.0 N
2	$A = \pi \times 15.0^2$	706.86 m ²
3	$2\rho A = 2 \times 1.225 \times 706.86$	1,731.80
4	$T / (2\rho A)$	147.2800
5	$v_i = \sqrt{(147.2800)}$	12.136 m/s
6	$P_{ideal} = T \times v_i$	3.095 MW
7	$P_{shaft} \approx P_{ideal} / 0.75$	4.127 MW

Quantity	Rev 23 (26 m)	Rev 25 (30 m)	Status
Disk area	530.93 m ²	706.86 m ²	CALC
Disk loading	48.97 kg/m ²	36.78 kg/m ²	CALC
P_{ideal}	≈3.57 MW	≈3.10 MW	CALC
$P @ FM 0.75$	≈4.76 MW	≈4.13 MW	CALC/STUDY

21.2 Mission Profile

Mission Parameter	Target Value	Status	Note
Design range	1,000 km	TARGET	Must be closed by mission fuel model
Crew + occupants	2 crew + 8 passengers	TARGET	8-person capsule limit
Fuel state in 26,000 kg budget	FULL usable	ASSUMPTION /STUDY	Rev 24/25 explicit FULL-fuel accounting

Mission Parameter	Target Value	Status	Note
Usable fuel planning mass	≈2,600 kg	ASSUMPTION /STUDY	Within 3,400 kg fuel-system line
Usable fuel volume ($\rho=0.80$ kg/L)	≈3,250 L	CALC/STUDY	Density/temperature compensation TBD
Cruise speed	TBD	OPEN	Depends on rotor/drag/engine selection
Cruise altitude	TBD	OPEN	Optimized for range/power balance
Reserve policy	TBD	OPEN	Per applicable civil/military requirement
IFR diversion reserve	TBD	OPEN	Typically 20–30 min additional fuel
Unusable fuel	~120 kg	PLANNING	Included in fuel budget
OEI diversion case	Required	REQ	Safety-case requirement
Hot/high performance	Required	REQ	ISA +20°C at 3,000 m minimum target

■ The 1,000 km range at 26,000 kg design mass with two turboshaft-class engines remains a programme TARGET. Reduced hover power at the 30.0 m rotor configuration does not by itself translate into a longer demonstrated range; cruise SFC, aerodynamic drag, rotor performance, and engine selection must still close before any range claim is made. Fuel quantity is held constant at the FULL 3,400 kg line specifically so the 26,000 kg mass accounting remains transparent and comparable across revisions.

22. Technology Readiness Levels

TRL assessment per the NASA/ESA TRL scale (1–9). Values are engineering judgments pending formal TRL review. The Rev 24/25 rotor-diameter change does not by itself advance any TRL; it changes the geometry and mass basis against which future evidence will be generated.

System / Subsystem	TRL Today	Basis for Assessment	Target G2	Target G8
Six-layer survival architecture concept	TRL 2	Architecture defined; no hardware	TRL 4	TRL 8
Deterministic Safety Kernel (IP-A)	TRL 2	Precondition set defined; no software	TRL 4	TRL 8
Capsule CFRP roll-cage concept	TRL 2–3	Similar CFRP capsule structures exist in spacecraft	TRL 5	TRL 8
Eight fixed landing supports	TRL 2	Similar energy-absorbers exist; capsule-integration novel	TRL 4	TRL 7
Stage S0–S2 (arming, fuel isolation)	TRL 3	Analogous systems exist in military jettison hardware	TRL 5	TRL 8
Stage S3 (capsule-to-airframe release)	TRL 1	No direct analog identified	TRL 3	TRL 7
3+2 canopy recovery	TRL 3	Three-main canopy architecture proven in spacecraft	TRL 5	TRL 7
Crashworthy seats (≥ 150 mm stroke)	TRL 3–4	Analogous seats exist in military helicopters	TRL 5	TRL 7
16-airbag capsule architecture	TRL 1–2	Airbag technology mature; capsule-integration novel	TRL 3	TRL 6
Sealed-capsule airbag gas management (OI-18)	TRL 1	No analogous system identified	TRL 3	TRL 6

System / Subsystem	TRL Today	Basis for Assessment	Target G2	Target G8
Mass/CG/inertia model	TRL 2	Planning model only, now at 30.0 m rotor basis	TRL 5	TRL 8
6-DOF separation dynamics model	TRL 1	No simulation started	TRL 4	TRL 8
30.0 m main rotor structural/dynamic design	TRL 1	Diameter set by CALC; no structural design started	TRL 3	TRL 7
IP-A through IP-D patent claims	TRL 1	Architecture defined; no legal filing	Prior art searched	Filed

PART VI

Geometry and Mass Accounting

Top-level mass tables at the 30.0 m rotor configuration, capsule component mass, detailed aircraft mass breakdown, mass-closure confidence, and preliminary capsule geometry

Top-Level Technical Baseline — Master Table

Parameter	Rev 25 Value	Status	Key Note
Design mass	26,000 kg	TARGET	Planning closure; bottom-up pending
Maximum capsule occupancy	8 persons	LOCKED LIMIT	Generation 1 hard limit
Main rotor diameter	30.0 m	TARGET	Rev 24/25 locked configuration
Main rotor disk area	706.86 m ²	CALC	$\pi \times (15.0)^2$
Disk loading	36.78 kg/m ²	CALC	See §4.1
Tail rotor diameter	4.0 m	TARGET	Anti-torque closure pending
Tail rotor disk area	12.57 m ²	CALC	$\pi \times (2.0)^2$
Capsule mass	7,000 kg	TARGET	Bottom-up pending Doc 113
Range target	1,000 km	TARGET	Mission model pending §21
Ideal hover power	≈3.10 MW	CALC	Actuator disk, ISA SL
Shaft power @ FM 0.75	≈4.13 MW	CALC	Study value only
Propulsion	Two turboshaft engines	REQ/STUDY	Engine selection open
Overall length	16.5–18.0 m	TARGET/STUDY	Layout reconciliation pending
Fuselage width (max)	4.0–4.5 m	TARGET/STUDY	
Overall height	4.8–5.3 m	TARGET/STUDY	
Main rotor separation	Not part of core safety case	DEFERRED	§12
Recovery canopies	3 primary + 2 reserve	STUDY	§11
Separation timeline	Staged S0–S5	TARGET/OPEN	§13
Occupant airbags	16 total (2 per occupant)	TARGET/STUDY	§10A; OI-18
Crashworthy seat stroke	≥150 mm dynamic stroke	TARGET/STUDY	§10B
Peak spinal load limit	≤6,670 N reference criterion	TARGET/STUDY	FAR 27.562 proxy; §10B
Main-rotor system mass	2,900 kg	ASSUMPTION/STUDY	+800 kg vs. Rev 23
Growth / configuration reserve	1,200 kg	PROGRAM RESERVE	–800 kg vs. Rev 23
Fuel system + FULL usable fuel	3,400 kg	ASSUMPTION/STUDY	Usable ≈2,600 kg / ≈3,250 L
Development gate	G0 approaching closure	CURRENT	G1–G8 not started

Mass Accounting — 26,000 kg (30.0 m Rotor Configuration)

Mass Group	Planning Mass	Status
Survival capsule, complete	7,000 kg	TARGET
Main airframe excluding capsule	5,000 kg	ASSUMPTION/STUDY
Main rotor, hub and controls	2,900 kg	ASSUMPTION/STUDY
Tail rotor, gearbox and drive	480 kg	ASSUMPTION/STUDY
Engines, transmission, gearboxes and mounts	3,100 kg	ASSUMPTION/STUDY
Fuel system + full usable fuel (FULL)	3,400 kg	ASSUMPTION/STUDY
Rapid separation system	1,000 kg	ASSUMPTION/STUDY
Avionics, electrical, hydraulic, thermal, control	1,600 kg	ASSUMPTION/STUDY
Operating fluids and consumables	320 kg	ASSUMPTION/STUDY
Growth / configuration reserve	1,200 kg	PROGRAM RESERVE
TOTAL	26,000 kg	CALC/ACCOUNTING CLOSURE

Capsule Component Mass — 7,000 kg (Unchanged by Rotor Configuration)

Capsule Component	Planning Mass	Notes
Primary crashworthy cell / internal roll-cage	1,950 kg	CFRP + titanium nodes
Outer capsule shell / fairing / thermal protection	500 kg	
Eight fixed landing supports + progressive energy absorbers	850 kg	
Eight crashworthy seats + restraints (≥ 150 mm stroke, $\leq 6,670$ N)	700 kg	Rev 20 corrected spec
Sixteen occupant airbags + inflators + sensors + wiring	200 kg	Rev 20 corrected (was 80–120 kg)
Emergency exits / doors, frames and seals	250 kg	
Distributed multi-canopy recovery system (3 primary + 2 reserve)	650 kg	
Independent capsule emergency power + batteries	250 kg	
Life support / emergency oxygen / environmental	200 kg	
Avionics / sensing / communications / locator / lighting	250 kg	
Separation interfaces + capsule-side release hardware	350 kg	
Internal secondary structure / floor / partitions / padding	300 kg	Reduced to absorb airbag mass
Fire / smoke detection and suppression	150 kg	
Water landing / flotation / sealing	150 kg	
Inspection / access / replaceable protection	100 kg	
Capsule configuration growth allowance	150 kg	

Capsule Component	Planning Mass	Notes
TOTAL	7,000 kg	CALC/CLOSURE

24. Preliminary Capsule Geometry Estimate

Shows geometric plausibility of 8 occupants + airbags + crashworthy seats in a capsule of 7,000 kg. All values pending Document 112 closure. Capsule geometry is independent of main rotor diameter.

Geometry Parameter	Preliminary Estimate	Status
Capsule outer diameter	≈ 3.2–3.6 m	STUDY
Capsule internal usable diameter	≈ 2.6–3.0 m (after roll-cage wall thickness)	STUDY
Capsule internal height (usable)	≈ 1.75–2.0 m (seated + headroom)	STUDY
Capsule outer shell thickness (CFRP + fairing)	≈ 100–150 mm per side	STUDY
Roll-cage member cross-section	≈ 60–100 mm diameter CFRP tubes	STUDY
Seat arrangement	2 rows × 4 seats, or circular/staggered 8-seat layout	OPEN
Seat pitch	≈ 650–750 mm	STUDY
Seat width	≈ 450–500 mm per position	STUDY
Seat stroke height occupied by absorber	≈ 150–200 mm below seat pan	STUDY

A circular capsule of ~3.4 m outer diameter, ~2.8 m internal diameter, ~1.85 m internal height accommodates 8 crashworthy seats in a 2×4 configuration with ≈700 mm pitch; 16 airbags stowed in headrests, sill-mounted and seat-back positions; 5-point restraints; emergency exits at a minimum of 2 locations with airbag-compatible egress paths; and internal volume for life support, emergency power, avionics and fire suppression. Geometric plausibility is supported at this preliminary level — bottom-up closure is required (Documents 112, 113).

25. Detailed Aircraft Mass Breakdown — Rev 25

Main Airframe — 5,000 kg (Unchanged)

Fuselage primary load-bearing structure 1,450 kg; frames, bulkheads and longerons 650 kg; tail boom and tail structural assemblies 520 kg; engine and transmission support structure 380 kg; rotor mast/transmission support integration 260 kg; external skins, fairings and aerodynamic panels 310 kg; doors, access panels and hinges 190 kg; firewalls, barriers and structural fire protection 170 kg; structural fittings, lugs, fasteners and local reinforcements 260 kg; cabin/service floor structure outside capsule boundary 210 kg; integration margin remainder to 5,000 kg. All values ASSUMPTION/STUDY pending bottom-up closure.

Main Rotor System — 2,900 kg (Rev 25 Planning Split)

Scaling basis: diameter ratio 30/26 ≈ 1.154; disk-area ratio ≈ 1.33. Blade and hub mass are expected to increase faster than linear with diameter. A planning uplift of +800 kg on the Rev 23 2,100 kg line is adopted as ASSUMPTION/STUDY until a rotor Bill of Materials exists (R24-OI-01).

Element	Mass	Status
Main blades (set)	1,450 kg	ASSUMPTION/STUDY
Hub, hinges/bearings, pitch links	650 kg	ASSUMPTION/STUDY

Element	Mass	Status
Swashplate / controls at hub	280 kg	ASSUMPTION/STUDY
Rotor mast / upper shaft allocation	220 kg	ASSUMPTION/STUDY
Blade fold/handling / local structure (if any)	100 kg	ASSUMPTION/STUDY
Integration / contingency within rotor line	200 kg	ASSUMPTION/STUDY
TOTAL	2,900 kg	ASSUMPTION/STUDY

Tail Rotor System — 480 kg (Unchanged)

Tail rotor blades (4.0 m) 145 kg; hub and blade retention 65 kg; tail rotor gearbox 95 kg; drive shafting, couplings and bearings 60 kg; controls and pitch mechanism 35 kg; tail structure/interface hardware + margin remainder. All values ASSUMPTION/STUDY pending bottom-up closure.

Engines and Transmission — 3,100 kg (Unchanged)

Engine #1 complete installed 700 kg; Engine #2 complete installed 700 kg; engine accessories and engine-control hardware 180 kg; main gearbox / combining transmission 650 kg; intermediate/reduction transmission elements 180 kg; drive shafts and couplings 100 kg; lubrication, oil cooling and filtration 80 kg; engine/gearbox mounts and vibration isolation 60 kg; exhaust and hot-section installation 50 kg; propulsion integration margin remainder. All values ASSUMPTION/STUDY pending bottom-up closure. The reduced hover-power demand at the 30.0 m rotor configuration (§21.1) does not by itself authorize a smaller engine class; see R24-OI-05.

Fuel System + Usable Fuel — 3,400 kg (FULL, Unchanged Line)

Element	Mass	Status
Usable fuel (FULL) — planning	2,600 kg	ASSUMPTION/STUDY
Tanks (crash/fire-resistant structure)	280 kg	ASSUMPTION/STUDY
Pumps / boost / transfer	90 kg	ASSUMPTION/STUDY
Lines, hoses, valves, fittings	110 kg	ASSUMPTION/STUDY
Filters, sensors, quantity measurement	55 kg	ASSUMPTION/STUDY
Isolation / emergency shutoff hardware	75 kg	ASSUMPTION/STUDY
Venting, surge, inerting / fire protection	70 kg	ASSUMPTION/STUDY
Unusable / residual fuel allocation	120 kg	ASSUMPTION/STUDY
TOTAL	3,400 kg	ASSUMPTION/STUDY

Usable volume at $\rho=0.80$ kg/L: $2,600 / 0.80 \approx 3,250$ L [CALC/STUDY]. Exact density, temperature compensation, and tank geometry remain TBD.

Avionics / Electrical / Hydraulic / Thermal — 1,600 kg (Unchanged)

Flight-control computers 180 kg; flight-control actuators/servos 210 kg; navigation/air-data/flight sensors 100 kg; communications/mission avionics 90 kg; electrical generation/distribution 190 kg; batteries/emergency electrical 90 kg; wiring/harnesses/connectors 170 kg; hydraulic pumps/reservoirs/lines/actuators 180 kg; thermal management/cooling 100 kg; environmental control/ventilation 70 kg; fire detection/warning/suppression interfaces 55 kg; health monitoring/data recording 45 kg; lighting/service/miscellaneous + margin 120 kg. All values ASSUMPTION/STUDY pending bottom-up closure.

39. Mass Closure Confidence Matrix

Mass Block	Rev 25 Planning	Current Basis	Closure Evidence	Tree Anchor	Impact
Complete survival capsule	7,000 kg	Planning	Bottom-up component database + weighed articles	MASS-003; capsule	High
Main airframe	5,000 kg	Assumption/study	Bottom-up structural and systems mass model	MASS-004	High
Main rotor / hub / controls	2,900 kg	Assumption/study	Rotor BoM + design definition (R24-OI-01)	MASS-005; ROT	High
Tail rotor / gearbox / drive	480 kg	Assumption/study	Component definition + supplier / design data	ROT / PROP	Medium
Engines / transmission / mounts	3,100 kg	Assumption/study	Selected engine and gearbox data + installation model	MASS-006; PROP	High
Fuel system + FULL usable fuel	3,400 kg	Assumption/study	Fuel accounting + tank/pump/line definition	MASS-012; FUEL	High
Rapid separation system	1,000 kg	Assumption/study	Mechanism and interface mass model	SEP domain	High
Avionics / electrical / hydraulic / thermal	1,600 kg	Assumption/study	Subsystem mass models + harness estimate	AV / ELEC	Medium
Operating fluids / consumables	320 kg	Assumption/study	Fluid accounting	MASS series	Low
Growth / configuration reserve	1,200 kg	Program reserve	Reserve policy + uncertainty tracking (reduced from 2,000 kg)	Mass control plan	High
Airbags (inside capsule 200 kg line)	200 kg	Corrected estimate	Module + ECU + harness + mounting bill of materials	CAP-AB docs	High
Crashworthy seats + restraints	700 kg	Planning line	Seat supplier data + stroke mechanism mass	CAP-SEAT docs	High

PART VII

Verification, Maturity and Evidence Architecture

Gate structure, evidence ladder, engineering maturity framework, engineering claim architecture, and subsystem TRL/evidence/gate matrix

26. Verification, Validation and Evidence Program

Gate	Evidence Objective	Key Evidence Items	Status
G0	Architecture Freeze	Safety philosophy; boundaries; interfaces; emergency states; airbag and seat concepts locked; 30.0 m rotor geometry locked	Approaching closure
G1	Analytical Closure	Mass, CG, inertia, rotor, power, fuel, structural load paths, separation kinematics, seat-stroke analysis	Not started
G2	Subsystem Demonstration	Release mechanisms, locks, absorbers, parachute, airbag module, seat stroke, sensors, independent power	Not started
G3	Integrated Ground Demonstration	End-to-end sequence; timing; state sensing; capsule clearance; airbag deployment in representative volume	Not started
G4	6-DOF Simulation	Vehicle and capsule dynamics; partial/asymmetric release; parachute inflation; sealed-capsule airbag gas; 30.0 m disk clearance	Not started
G5	Structural / Crash Testing	Capsule, attachments, seats (dynamic stroke test per FAR 27.562 proxy), airbag sled tests, supports, energy absorbers	Not started
G6	Recovery System Testing	Canopy deployment; inflation loads; failure cases; stability; landing performance	Not started
G7	Flight Development	Progressive envelope expansion; normal and emergency recovery validated in flight	Not started
G8	Certification / Qualification	Full evidence package against selected certification basis	Not started

■ Terminology: Specified = defined in design; Analyzed = documented analytical result; Simulated = supported by a model; Tested = demonstrated in a defined test; Verified = evidence shows compliance; Validated = suitability established for intended use; Qualified/Certified = formal processes complete. As of Rev 25: all HSH-00 claims are Specified or Studied. None are Analyzed, Simulated, Tested, Verified, Validated, Qualified, or Certified.

27. Engineering Maturity Framework

Configuration Item	Rev 25 Control Rule	Closure Evidence Required
MTOW / design mass	Freeze only after bottom-up mass model and reserve policy reconciled	Mass report + uncertainty analysis
Rotor geometry	Maintain 30.0 m target until performance study closes	Rotor sizing + power model + R24-OI-01/02
Capsule mass	7,000 kg planning target; no closure claim	Bottom-up mass budget (Doc 113)
Occupancy	8 persons maximum — LOCKED for Generation 1	Configuration control only

Configuration Item	Rev 25 Control Rule	Closure Evidence Required
Recovery	3 primary + 2 reserve study architecture	Sizing + inflation dynamics + failure testing
Separation	Non-rotor candidate set only in core baseline	6-DOF + clearance + ground demonstration
Safety authorization	Deterministic kernel remains authoritative	Requirements + software/hardware verification
Airbags	16 total (2 per occupant) TARGET/STUDY	Packaging + sled/impact tests + OI-18 gas analysis
Crashworthy seat stroke	≥150 mm stroke; ≤6,670 N criterion	Dynamic seat tests + analysis

Maturity roadmap: Architecture control (G0) → Analytical closure (G1) → Subsystem design (G2 prep) → Ground demonstration (G3) → Model validation (G4) → Structural/recovery tests (G5/G6) → Flight development (G7) → Qualification/certification (G8).

32. Engineering Claim Architecture

Purpose: convert the principal HSH engineering and IP propositions into traceable engineering claims. Each claim is connected to the evidence that must exist before the claim can be treated as demonstrated. This is an architecture for closure, not a statement that the claims are already proven.

ID	Engineering Claim	Primary Evidence Required	Verification Path	Basis
EC-00 1	Occupied survival capsule can become the final landing boundary in the defined emergency architecture	Requirements, structural load paths, attachment analysis, separation dynamics, clearance evidence	Analysis → 6-DOF simulation → ground demonstration → applicable testing	Specified / Studied
EC-00 2	State-dependent selective separation can release designated non-rotor hazards while preserving capsule survivability	Release mechanism design, authorization logic, partial/asymmetric cases, failed-release fallback	Analysis → mechanism test → integrated ground demonstration	Specified / Open
EC-00 3	Deterministic Safety Kernel can remain the authoritative authorization boundary for safety-critical actuation	Requirements, precondition logic, independence analysis, formal/robustness evidence	Requirements verification → formal methods / fault injection → qualification evidence	Specified / Open
EC-00 4	Eight capsule-integrated landing supports can provide progressive impact attenuation with asymmetric failure management	Load cases, FEA, absorber characteristics, terrain cases, failed-support redistribution	Analysis → drop testing → integrated crashworthiness evidence	Specified / Open
EC-00 5	3 primary + 2 reserve canopy architecture can recover the 7,000 kg capsule within a validated deployment envelope	Sizing, inflation dynamics, failure tolerance, minimum altitude envelope	Sizing analysis → deployment tests → envelope closure	Study / Open
EC-00 6	16-airbag + ≥150 mm stroke seat architecture can limit occupant loads without defeating egress or sealed-capsule environment	Packaging, deployment logic, sled/impact tests, sealed-volume gas analysis (OI-18)	Design → sled/impact → gas analysis → combined attenuation evidence	Target / Open
EC-00 7	IP-A through IP-D remain coherent conceptual claim surfaces supported by the engineering architecture	Prior-art posture, claim definitions, engineering traceability	Professional prior-art search → claim drafting → legal review	Conceptual / Not filed

ID	Engineering Claim	Primary Evidence Required	Verification Path	Basis
EC-008	30.0 m main rotor can be structurally and dynamically closed at the 2,900 kg planning mass line	Rotor BoM, blade/hub design, dynamic clearance, ground resonance analysis	Design → structural analysis → dynamic testing (R24-OI-01/02)	Target / Open

37. Evidence Ladder

All HSH-00 claims are Specified or Studied. The ladder is the controlled progression from specification to demonstrated evidence.

Ladder Step	Meaning	Minimum Evidence Content	Typical Gate
E0	Specified / Studied	Requirement or concept written with status label	HSH-00 (current)
E1	Analyzed	Documented analytical result with stated assumptions and uncertainty	G1 mass, loads, kinematics
E2	Simulated	Model result with stated fidelity and validation status	G4 6-DOF; OI-18 gas model
E3	Subsystem tested	Defined test article, conditions, instrumentation, results	G2 mechanisms, seats, airbags
E4	Integrated demonstrated	End-to-end ground sequence under controlled conditions	G3 integrated demonstration
E5	Flight / qualification evidence	Flight development and formal qualification against basis	G7–G8

38. Subsystem TRL + Evidence + Gate Matrix

Extends the TRL table by adding the evidence needed to move each subsystem toward programme gates. TRL values are not re-scored upward by documentation alone.

Subsystem	TRL Today	Evidence Needed	Next Gate	Target Evidence
Six-layer survival architecture	2	Independence analysis, CCA, fault injection	G1–G3	E1–E4
Deterministic Safety Kernel (IP-A)	2	Requirements, formal/robust verification, independence from AI	G1–G2	E1–E3
Capsule CFRP roll-cage	2–3	FEA, joint/node design, crash load cases	G5	E1–E3
Eight fixed landing supports	2	Absorber characterization, drop tests, asymmetric cases	G5	E2–E4
Stage S0–S2 (arming, fuel isolation)	3	Mechanism tests, timing, isolation verification	G2–G3	E3–E4
Stage S3 (capsule-to-airframe release)	1	Mechanism design, 6-DOF, ground demonstration	G2–G4	E1–E4
3+2 canopy recovery	3	Sizing, inflation, failure cases, min altitude	G6	E2–E4
Crashworthy seats ≥150 mm / ≤6,670 N	3–4	Dynamic seat tests (FAR 27.562 proxy)	G2–G5	E3–E4
16-airbag capsule architecture	1–2	Packaging, deployment, sealed-volume gas, sled tests	G2–G5	E2–E5

Subsystem	TRL Today	Evidence Needed	Next Gate	Target Evidence
Mass / CG / inertia model	2	Bottom-up mass database, uncertainty, CG/inertia closure	G1	E2
6-DOF separation dynamics	1	Model specification, correlation plan	G4	E3
30.0 m main rotor structural/dynamic design	1	Rotor BoM, blade/hub design, dynamic clearance, ground resonance	G1–G5	E1–E4
IP-A through IP-D	1	Prior-art search, invention disclosures, claim drafting, legal review	External IP gates	E0–E1 legal

PART VIII

Master Configuration, Philosophy and Engineering Basis

Master configuration record, design philosophy, predecessor documents, external references, and the final Generation 1 baseline statement

28. Master Configuration Record

Configuration Item	Rev 25 Baseline	Status
Aircraft programme name	HSH Heavy Safe Helicopter	LOCKED
Founder	Amirbahador Atai	CONTROLLED
Generation	Generation 1	LOCKED
Document number	HSH-00	LOCKED
Revision	Rev 25 — Professional Master Edition (30.0 m Rotor Configuration)	BASELINE
Capsule occupancy maximum	8 persons	LOCKED LIMIT
Vehicle design mass	26,000 kg (FULL fuel)	TARGET/OPEN
Capsule mass allocation	7,000 kg	PLANNING/OPEN
Main rotor target	30.0 m	TARGET/OPEN
Tail rotor target	4.0 m	TARGET/OPEN
Range target	1,000 km	TARGET/OPEN
Landing supports	8 fixed energy-absorbing	CONCEPT/OPEN
Recovery	3 primary + 2 reserve canopies	STUDY/OPEN
Occupant airbags	16 (2 per occupant)	TARGET/STUDY
Crashworthy seat stroke	≥150 mm; ≤6,670 N	TARGET/STUDY
Core separation	Selective, staged, non-rotor only	CONCEPT/OPEN
Main-rotor separation	Excluded from core safety case	DEFERRED RESEARCH
Safety authorization	Deterministic Safety Kernel	CONCEPT/OPEN
IP claims	IP-A, IP-B, IP-C, IP-D	CONCEPTUAL/NOT FILED
Main-rotor system mass	2,900 kg	ASSUMPTION/STUDY
Growth / configuration reserve	1,200 kg	PROGRAM RESERVE
Development gate	G0 approaching closure	CURRENT

40. Configuration Baseline Diagram (Textual)

Textual configuration map of the locked and target elements. Visual CAD is out of scope for this document.

Configuration Class	Content	Control Rule
LOCKED LIMIT	8 persons maximum occupancy	Any change → full re-analysis
LOCKED CONCEPT	No retractable landing gear; capsule central; main rotor not in core separation set; 30.0 m diameter is the controlling geometry	Impact assessment required to alter
TARGET geometry	30.0 m main rotor; 4.0 m tail rotor; 26,000 kg design mass; 7,000 kg capsule	Bottom-up and performance closure
TARGET survival hardware	8 fixed energy-absorbing supports; 16 airbags; ≥150 mm stroke seats ≤6,670 N; 3+2 canopies	Design + test evidence
NORMAL FLIGHT LOAD PATHS	Airframe ↔ CAP-AF ↔ capsule structure ↔ seats	Primary flight loads
EMERGENCY LOAD PATH / FINAL LANDING BOUNDARY	Released capsule ↔ 8 supports ↔ crashworthy seats ↔ airbags ↔ occupants	Post-separation impact
CONTROLLED INTERFACES	CAP-AF, CAP-PAR, CAP-LG, CAP-AB, CAP-SEAT, JET-*, FUEL-ISO	ICD set under configuration control
STATE-DEPENDENT TRANSITION	Normal → Anomaly → Confirm → Assess → Recover if possible → Authorize → Selective separation → Clearance → Recovery → Landing	Emergency state machine
INDEPENDENCE PRINCIPLE	Physical, electrical and software segregation where required; human authority preserved; AI advisory only at the safety boundary	Non-negotiable

■ Configuration-control anchor: any change to occupancy, mass, rotor geometry, separation architecture, recovery architecture, landing-support count/geometry, airbag count/placement, seat stroke or Safety Kernel preconditions triggers the impact-assessment process already defined in this baseline. Rev 24/25 is itself such a change for main rotor diameter, disk area, disk loading, hover power, main-rotor mass, and growth reserve.

41. Document Dependency / Parent–Child Architecture

The Master Engineering Document Tree contains requirements, subsystem design, safety, dynamics, verification, certification, IP and programme-management documents. This section adds a dependency discipline so that each lower-level document can identify its controlling parent, upstream inputs, downstream verification artifact and closure gate.

Document Layer	Example Parent	Example Child / Output	Dependency Rule
HSH-00 Master Baseline	HSH-00 Rev 25	REQ / SAF / MASS / ROT / SEP / REC / AV / DYN / VER / IP domains	Master baseline controls architecture, terminology, locked limits, 30.0 m rotor geometry and gate definitions
System Requirements	REQ-001–040	Subsystem requirements, derived requirements, RVM	Child requirements shall trace upward to parent REQ and to HSH-00
Safety domain	SAF plans + FHA/PSSA/SSA	CCA, PRA, ZSA, Safety Kernel verification	No safety claim without upward trace to HSH-00 layers and OI register
Capsule domain	CAP requirements + design	Roll-cage, seats, airbags, supports, mass, tests	Must respect 8-person lock, 7,000 kg allocation, OI-16/17/18
Rotor domain	ROT requirements + design	30.0 m rotor BoM, structural design, dynamic clearance	Must respect R24-OI-01/02 and 2,900 kg mass line
Separation domain	SEP requirements + design	Mechanisms, S0–S5, 6-DOF, ground demo	Main rotor remains excluded from core candidate set

Document Layer	Example Parent	Example Child / Output	Dependency Rule
Recovery domain	REC / PAR	3+2 sizing, deployment, envelope	OI-02 minimum altitude is critical path
Verification domain	VER plans + matrices	G0–G8 evidence packages	Evidence ladder E0–E5 applies
IP domain	IP-A/B/C/D descriptions	Prior-art search, claim drafts, FTO	No novelty assertion without external legal work

29. Engineering Basis and External References

Predecessor documents. LSH-00 Rev 3.2 (six-layer philosophy; independent initiation; controlled separation; eight supports; capsule-as-final-boundary concept); USCC Doc 00 Rev 1.3 (jettison sequence; deterministic safety authorization); HSC-00/HSR-00 Rev 12–24 (progressive engineering development, correction and rotor reconfiguration).

External engineering references. EASA CS-29 Amendment 12: large-rotorcraft certification standard; jettisonable configurations require clear trajectory, no damaging interaction, acceptable post-jettison stability and performance. FAR 27.562 / CS-27 dynamic seat-test requirements: proxy reference criterion for $\leq 6,670$ N peak spinal compressive load used in the crashworthy-seat specification (§10B). MIL-STD-1290A: helicopter crashworthiness design criteria; relevant to capsule structural design, seat design, landing-support design and occupant injury criteria. NASA TRL scale: used for TRL assessment in §22. Momentum theory (actuator disk) for ideal induced power at the 30.0 m rotor configuration; aviation kerosene density ≈ 0.80 kg/L used as a volume ASSUMPTION.

30. Design Philosophy — Controlled Survival, Not a Single Device

HSR Generation 1 integrates multiple survival mechanisms into one state-aware architecture. The vehicle does not depend solely on a parachute, solely on autorotation, solely on a detachable cabin, or solely on automatic software. Removing main-rotor separation from the core safety case — introduced in Rev 13 and retained through Rev 25 — strengthens rather than weakens the layered proposition: a safety case that requires its riskiest, least-proven capability to succeed is not actually layered. Rev 25 extends this philosophy by relieving disk loading through a larger 30.0 m rotor rather than by increasing design mass, paying the mass cost from programme reserve under transparent configuration control. Documentation is not a substitute for evidence. The path from this baseline to G8 is long and requires real engineering work — but the architecture is defined clearly enough, honestly enough, and with sufficient IP specificity to serve as a credible starting point for that work.

31. Final Generation 1 Baseline Statement

HSR-00 Rev 25 is the definitive Generation 1 conceptual master baseline for the HSR Heavy Safe Helicopter at the 30.0 m main-rotor configuration. It is the document a technical due-diligence team from a potential IP licensee should receive as the starting point for evaluation.

What is defined: the full six-layer survival architecture; the Safety Kernel and its precondition set; the staged S0–S5 separation timeline with an honest confidence-basis assessment; the 3+2 canopy recovery architecture with preliminary sizing; the 8-point capsule-integrated landing-support system; the 16-airbag and crashworthy-seat occupant-protection architecture (with corrected engineering specifications); four IP claims (IP-A through IP-D) with prior-art awareness; a full open-items register (OI-01 through OI-18, plus R24-OI-01 through R24-OI-05); a TRL assessment; a competitive-differentiation table; a preliminary capsule geometry check; a mission-profile framework; the 30.0 m rotor geometry, disk loading and hover-power study; and the diligence enhancement layer of Part IX.

What remains to be proven: everything in the open-items register. The gap between architecture and evidence is the scope of the Generation 1 engineering document tree (Documents 001–500).

PART IX

Safety-Case Synthesis and Failure Containment

Layered safety-case argument, failure-containment philosophy, and Rev 24/25 rotor-configuration open items

35. Safety Case Architecture

The safety case is organized as a layered argument: the vehicle shall provide multiple independent means of protecting occupants, ordered from conventional recovery to capsule separation and post-landing survival. The safety case does not rely on main-rotor separation. Each layer requires defined requirements, hazards, verification methods and residual-risk acceptance.

Safety-Case Block	Argument	Evidence Required	Principal Residual Risk
Layer L1–L3 initiation	Automatic primary, automatic backup, and non-bypassable crew authority	Independence analysis, fault injection, HMI verification	Common-cause defeat of all initiation paths
Layer L4 selective separation	State-dependent release of designated non-rotor hazards only	Mechanism design, S0–S5 timing, 6-DOF, ground demonstration	Unintended release; incomplete/asymmetric release
Layer L5 direct-impact survival	Capsule structure, roll-cage, 8 supports, crashworthy seats, 16 airbags	FEA, drop/sled tests, OI-18 gas analysis, injury criteria	Bottom-out; overpressure; egress blockage
Layer L6 recovery / post-event	3+2 canopies, independent power, locator, life support	Sizing, deployment tests, minimum altitude envelope	Low-altitude non-availability; canopy failure
Safety Kernel boundary	Deterministic authorization of safety-critical actuation	Requirements, formal/robust verification, independence from AI	AI bypass; incomplete preconditions
Rotor configuration boundary	30.0 m rotor kept attached through the entire survival chain	Structural/dynamic analysis, clearance study (R24-OI-02)	Larger rotor increases collision/clearance consequence if ever separated

36. Failure Containment Philosophy

The architecture identifies unintended separation, partial/asymmetric release, sensor loss, power-bus failure, AI/ML failure, fire during separation, released-component collision, parachute out-of-envelope deployment, landing-support failure, airbag inadvertent/non-deployment, sealed-capsule overpressure/toxicity (OI-18), and seat-stroke bottom-out. Containment philosophy: prevent common-cause defeat of multiple layers; detect degraded states; define a safe fallback for every failed-release configuration; preserve human authority; keep AI advisory at the safety boundary.

Failure Class	Containment Design Response	Program Rule
Unintended separation	Mechanical locks + electrical inhibits + command validation + monitoring	Quantitative safety objective + CCA
Partial / asymmetric release	State sensing + asymmetric-load analysis + degraded recovery mode	First-class emergency state, not an afterthought

Failure Class	Containment Design Response	Program Rule
Sensor loss / bad data	Validity monitoring + diverse sensing + safe-state logic	Kernel must fail safe
Power-bus failure	Independent survival power for critical functions	No single bus defeats L1–L6 critical path
AI/ML failure	Deterministic Safety Kernel remains authoritative	Non-negotiable certification boundary
Sealed-capsule airbag gas (OI-18)	Sealed-volume analysis + staged/vented deployment options	Must close before deployment-logic freeze
30.0 m rotor clearance after release (R24-OI-02)	6-DOF clearance analysis at enlarged disk geometry	Must close before S3/S4 timeline is treated as validated

42. Rev 24/25 Rotor-Configuration Open Items

ID	Item	Status
R24-OI-01	Bottom-up main-rotor BoM vs 2,900 kg planning line	OPEN
R24-OI-02	30.0 m rotor dynamics, clearance, ground resonance	OPEN
R24-OI-03	Cruise energy model for 1,000 km with FULL fuel	OPEN
R24-OI-04	Align HSH-001/002/003 and lower tree documents to Rev 25 30.0 m figures	TBD — DEPENDENCY
R24-OI-05	Installed power class vs reduced hover demand	STUDY/OPEN

Appendix A — Glossary of Abbreviations

Abbreviation	Meaning
CFRP	Carbon-fibre-reinforced polymer
CG	Center of gravity
CCA	Common-cause analysis
DOF	Degrees of freedom (6-DOF: full rigid-body translational + rotational dynamics)
ECU	Electronic control unit
FEA	Finite element analysis
FM	Figure of merit (rotor hover efficiency)
FMEA	Failure modes and effects analysis
FTO	Freedom to operate
ICD	Interface control document
IP	Intellectual property
KE	Kinetic energy
MTOW	Maximum takeoff weight
OEI	One engine inoperative
OI	Open item
PSSA/SSA/FHA	Preliminary system safety assessment / system safety assessment / functional hazard assessment
RVM	Requirements verification matrix
SFC	Specific fuel consumption
TRL	Technology readiness level (NASA/ESA 1–9 scale)
ZSA	Zonal safety analysis

Appendix B — Status Label Quick Reference

See Part I, Section 5.1 for the full precise definitions. This is a one-line quick reference for reviewers scanning tables.

Label	One-line meaning
LOCKED LIMIT / LOCKED CONCEPT	Fixed for Generation 1; formal change process required to alter
REQ	Binding requirement unless formally waived
TARGET	Development objective, not demonstrated
ASSUMPTION / STUDY	Working value pending analysis or test
CALC	Calculated from stated assumptions/targets
OPEN / OPEN — CRITICAL	Requires closure; critical items sit on the P0 path
DEFERRED — RESEARCH TRACK	Excluded from the licensable core baseline

Appendix C — Rev23 → Rev25 Configuration Delta Summary

A single consolidated reference for reviewers who already know Rev 23 and need only the delta.

Parameter	Rev 23	Rev 25
Main rotor diameter	26.0 m	30.0 m
Main rotor radius	13.0 m	15.0 m
Disk area	530.93 m ²	706.86 m ²
Disk loading	48.97 kg/m ²	36.78 kg/m ²
Ideal hover power	≈3.57 MW	≈3.10 MW
Shaft power @ FM 0.75	≈4.76 MW	≈4.13 MW
Main rotor / hub / controls mass	2,100 kg	2,900 kg
Growth / configuration reserve	2,000 kg	1,200 kg
Aircraft design mass	26,000 kg	26,000 kg (FULL fuel, explicit)
Fuel system + usable fuel line	3,400 kg	3,400 kg (usable ≈2,600 kg / ≈3,250 L)
Capsule mass	7,000 kg	7,000 kg (unchanged)
Occupancy	8 (LOCKED)	8 (LOCKED, unchanged)
Range target	1,000 km	1,000 km (unchanged)
Open items	OI-01–OI-18	OI-01–OI-18 + R24-OI-01–05
Document length / depth	46-page professional master edition	Full master-edition depth restored at 30.0 m configuration

END OF HSH-00 REVISION 25 — PROFESSIONAL MASTER EDITION

Full Rev 23 structure, depth, tables and open-items register carried forward | Rev 24 30.0 m rotor configuration reconciled throughout | Founder: Amirbahador Atai

Not a certification approval | Not a filed patent | Not demonstrated performance

HSH HEAVY SAFE HELICOPTER — GENERATION 1

HSH-00 Rev 26 — Integrated Professional Master Edition

33.0 m Main Rotor / 28,000 kg Planning Configuration / Expanded 21-Feature Survival Architecture

Founder: Amirbahador Atai

Status: Conceptual Engineering Baseline — Pre-Simulation / Pre-Hardware / Pre-Prototype

CONTROLLED REV26 STATEMENT

Rev26 preserves the complete Rev25 source document as controlled baseline content and adds a reconciled Rev26 change set covering the 33.0 m main rotor, the 28,000 kg planning mass target, and the 21 additional safety / recovery / verification features defined in this revision.

Mass-control note: the 28,000 kg figure is TARGET/OPEN. The calculated midpoint incremental programme mass is approximately 2021 kg, including an estimated 780 kg rotor-system uplift. Bottom-up mass closure remains mandatory.

1. REV26 CONFIGURATION CHANGE — 30.0 m → 33.0 m MAIN ROTOR

The main-rotor diameter changes from 30.0 m to 33.0 m. The purpose is to reduce disk loading and retain a comparable hover-power study basis while the planning mass increases toward 28,000 kg. This is a configuration trade, not evidence that the added mass has already been structurally or dynamically closed.

Parameter	Rev25	Rev26	Status / basis
Design mass	26,000 kg	28,000 kg	TARGET / OPEN
Main rotor diameter	30.0 m	33.0 m	TARGET / OPEN
Rotor disk area	706.86 m²	855.30 m²	CALC
Disk loading	36.78 kg/m²	32.74 kg/m²	CALC
Induced velocity	12.136 m/s	11.449 m/s	CALC
Ideal hover power	≈3.10 MW	≈3.14 MW	CALC / STUDY
Shaft power @ FM 0.75	≈4.13 MW	≈4.19 MW	CALC / STUDY
Main rotor / hub / controls	2,900 kg	≈3680 kg	ASSUMPTION / STUDY
Midpoint programme increment	—	≈+2021 kg	OPEN / closure required

At 28,000 kg and 33.0 m, the study disk loading is 32.74 kg/m². Ideal hover power is approximately 3.14 MW and shaft power at FM 0.75 approximately 4.19 MW. Engine sizing, rotor structural design, dynamics, clearance, ground resonance and OEI performance remain OPEN.

2. REV26 MASS INTEGRATION

Mass item	Planning increment	Status
33 m rotor-system uplift vs Rev25	≈+780 kg	ASSUMPTION / STUDY
F01–F18 physical feature midpoint	≈1241 kg	PLANNING / OPEN
F19–F21 engineering-only features	0 kg aircraft hardware allocation	PROGRAMME / OPEN
Combined midpoint increment	≈2021 kg	OPEN — bottom-up closure
26 t baseline + midpoint increment	≈28021 kg	INDICATIVE ONLY
Rev26 planning configuration	28,000 kg	TARGET / OPEN

Planning range for F01–F18 is approximately 852–1630 kg. The broad range is deliberate because packaging, propellant, certification hardware, structural integration and redundancy are not yet designed. The 28,000 kg target is therefore a controlled planning target, not a closed MTOW.

3.1 REV26 — NEW SURVIVAL / RECOVERY / VERIFICATION FEATURES

ID	Feature	Mass kg	Mid	Engineering integration
F01	Four micro-jet capsule attitude-control system (CPSACS)	340–565	~450	Post-separation pitch/roll/yaw and attitude conditioning; deterministic authorization; 6-DOF closure.
F02	Independent capsule recovery computer	15–30	~23	Dedicated recovery-state computation independent of the primary aircraft flight computer.
F03	Independent capsule IMU / altimeter set	20–35	~28	Independent inertial/altitude sensing for post-separation control and recovery gating.
F04	Capsule health monitoring system	15–30	~23	Monitoring of capsule power, pressure, temperature, structure, recovery and life-support state.
F05	Capsule recovery-readiness state	5–15	~10	Deterministic readiness state for separation, attitude control, canopy and landing-support availability.
F06	Anti-spin drogue / stabilization device	30–60	~45	Limits post-release angular-rate growth and conditions the capsule for recovery deployment.
F07	Advanced / controlled canopy architecture	50–120	~85	Controlled or staged canopy deployment; final architecture OPEN.

F01–F18 remain subordinate to the deterministic Safety Kernel. None independently authorizes capsule release, canopy deployment, occupant-protection deployment or other safety-critical actuation outside the defined authorization chain.

3.2 REV26 — NEW SURVIVAL / RECOVERY / VERIFICATION FEATURES

ID	Feature	Mass kg	Mid	Engineering integration
F08	Water-survival / flotation system	100–180	~140	Capsule buoyancy, sealing and post-ditching stability for water-landing scenarios.
F09	Fire-isolated capsule architecture	80–150	~115	Thermal/fire barriers and protection of survival-critical systems.
F10	Smart separation interfaces	40–80	~60	Instrumented release interfaces with state confirmation, monitoring and deterministic interlocks.
F11	Partial-separation control	10–25	~18	Explicit management of incomplete/asymmetric release states and degraded recovery modes.
F12	Terrain-relative recovery support	20–40	~30	Terrain/altitude-relative recovery logic constrained to validated envelopes.
F13	Capsule self-righting system	50–100	~75	Post-impact or water-entry attitude correction; architecture OPEN.
F14	Adaptive crash-pulse management	50–120	~85	Adaptive energy-absorption/load-path management across supports, seats and restraints.

Recovery and impact enhancements require 6-DOF dynamics, structural loads, occupant injury criteria, flotation/ditching analysis and validated deployment envelopes.

3.3 REV26 — NEW SURVIVAL / RECOVERY / VERIFICATION FEATURES

ID	Feature	Mass kg	Mid	Engineering integration
F15	Structural health monitoring	10–30	~20	Distributed structural/load monitoring for capsule and critical attachment/support elements.
F16	Independent emergency communications	10–25	~18	Independent post-separation emergency communications and rescue coordination.
F17	Independent emergency recorder	5–15	~10	Independent event recorder preserving sensor, state, authorization and recovery data.
F18	Automatic post-landing health report	2–10	~6	Automatic logging/transmission of capsule status and recovery condition.
F19	Survival-envelope / state-machine architecture	0	0	Controlled software/state architecture; no incremental aircraft mass assigned.
F20	HSH digital twin / 6-DOF model	0	0	Engineering model and simulation framework; no aircraft hardware mass assigned.
F21	Demonstrator / verification architecture	0	0	Defined test articles, rigs, instrumentation and integrated demonstration plan.

F19–F21 are architecture and verification enablers. Documentation or modelling alone does not increase TRL or constitute proof of performance.

4. REV26 UPDATED SAFETY ARCHITECTURE

Layer	Rev25 baseline	Rev26 enhancement	Control principle
L1 Initiation / sensing	Automatic primary, backup and crew authority	F03 independent sensing + F04 health monitoring + F15 structural monitoring	Diverse sensing; validity monitoring
L2 State estimation	Aircraft/capsule state assessment	F19 survival state machine + F20 digital twin / 6-DOF model	Model supports; deterministic logic authorizes
L3 Risk / recovery prediction	Emergency-state assessment	F12 terrain-relative recovery + F05 readiness state	Validated envelope only
L4 Selective separation	S0–S5 non-rotor separation	F10 smart interfaces + F11 partial-separation control	State-dependent; no main-rotor separation
L5 Direct-impact survival	8 supports + seats + 16 airbags	F14 adaptive crash-pulse management + F13 self-righting	Progressive load management
L6 Recovery / post-event	3+2 canopies + independent power/life support	F01 attitude control + F06 drogue + F07 controlled canopy + F08 flotation + F09 fire isolation + F16–F18	Independent capsule recovery, bounded by Kernel

Non-negotiable: AI/ML remains advisory at the safety boundary. Confidence is not authorization. The deterministic Safety Kernel remains the final authorization boundary for safety-critical actuation. Human emergency authority is preserved.

5. REV26 EMERGENCY STATE MACHINE EXTENSION

State	Rev26 entry	Primary action	Features
7A — CAPSULE CONTROL INITIATED	Capsule release authorized/detected	F03 sensing; F02 recovery computer; initialize F19	F02/F03/F05/F19
7B — ATTITUDE CONDITIONING	Rates/attitude outside recovery envelope	F01 micro-jets + F06 anti-spin; inhibit canopy until criteria met	F01/F06/F20
7C — RECOVERY READINESS	Attitude/rates/altitude/clearance valid	Confirm F05 readiness and authorize recovery	F05/F07/F12
7D — CONTROLLED RECOVERY	Primary recovery deployment authorized	F07 canopy; monitor F04/F15; terrain-relative state	F04/F07/F12/F15
7E — DEGRADED RECOVERY	Partial/asymmetric release or recovery fault	F11 degraded mode; reserve/fallback path	F11/F19
7F — WATER / TERRAIN LANDING	Water or hazardous terrain predicted/detected	F08 flotation; F13 self-righting; F14 impact management	F08/F13/F14
8 — POST-LANDING SURVIVAL	Capsule stabilized	F16 communications; F17 recorder; F18 health report	F16/F17/F18

This state-machine extension is conceptual. Timing, transition criteria, sensor tolerances, actuator authority, recovery envelopes and failure handling require simulation, subsystem tests and integrated ground demonstration.

6. REV26 UPDATED INTERFACE / HAZARD MATRIX

Hazard / failure state	Rev25 response	Rev26 added containment
Asymmetric capsule separation	State sensing + 6-DOF required	F10 interfaces + F11 degraded state + F01 attitude control
High post-separation angular rate	Not explicitly controlled	F01 micro-jets + F06 anti-spin; canopy gated by validated envelope
Bad capsule sensor data	Validity monitoring	F03 independent sensing + F04 health monitoring + deterministic cross-checks
Terrain unsuitable for landing	Generic recovery assessment	F12 terrain-relative recovery + F13 self-righting where applicable
Water landing	Flotation/sealing target	F08 dedicated flotation / water-survival architecture
Fire reaches survival-critical systems	Fire containment	F09 fire-isolated capsule architecture
Structural degradation	No dedicated SHM feature	F15 structural health monitoring
Recovery computer failure	Primary architecture	F02 independent recovery computer with defined fallback
Data loss after event	Post-event audit	F17 recorder + F18 health report
Canopy deployment instability	3+2 architecture	F01/F06/F07 coordinated attitude and deployment conditioning
Impact pulse exceeds design	Supports + seats + airbags	F14 adaptive crash-pulse management; dynamic-test closure

7. REV26 UPDATED VERIFICATION AND EVIDENCE PROGRAM

Gate	Rev26 evidence added	Primary features
G1 Analytical Closure	33 m rotor sizing; mass/CG/inertia; 6-DOF model; attitude-control authority; flotation and crash-pulse models	F01/F03/F08/F14/F20
G2 Subsystem Demonstration	Micro-jet bench test; recovery computer; sensing; smart interfaces; drogue; flotation; fire isolation; SHM; recorder/comms	F01–F04/F06/F08–F10/F15–F17
G3 Integrated Ground Demonstration	Staged sequence; partial/asymmetric separation; attitude conditioning; recovery readiness; representative occupant protection	F01/F05/F10/F11/F19/F21
G4 6-DOF Simulation	Vehicle/capsule dynamics; asymmetric release; rotor clearance; thruster control; drogue; canopy; terrain-relative logic	F01/F06/F12/F20
G5 Structural / Crash Testing	Capsule, supports, seats, airbags, adaptive pulse, fire isolation and self-righting where applicable	F08/F09/F13/F14/F21
G6 Recovery Testing	Canopy deployment/inflation, attitude control, anti-spin, reserve recovery, water survival	F01/F06/F07/F08/F13
G7 Flight Development	Progressive emergency-envelope validation	Applicable flight hardware
G8 Qualification / Certification	Formal compliance evidence against selected basis	Programme-level

Specifying these tests does not mean they have been performed. Rev26 remains conceptual until evidence is generated.

8. REV26 UPDATED IP / ENGINEERING VALUE SURFACES

Existing surface	Rev26 supporting additions	Evidence required
IP-A — Deterministic Safety Kernel	F02/F03/F05/F10/F11/F19	Requirements, independence, fault injection, formal/robust verification
IP-B — Eight-point capsule support + asymmetric failure	F13/F14/F15	FEA, dynamic tests, self-righting and adaptive load-path validation
IP-C — State-dependent non-rotor selective separation	F01/F03/F06/F10/F11/F12/F19/F20	6-DOF dynamics, mechanism tests, clearance and degraded-state evidence
IP-D — Integrated multi-layer survival chain	F01–F18 + F19–F21	Integrated system demonstration and recovery-envelope evidence

No Rev26 text asserts novelty, patentability, freedom-to-operate, certification, or demonstrated safety performance. Professional prior-art search and legal review remain mandatory.

9. REV26 MASTER CONFIGURATION RECORD

Configuration item	Rev26 baseline	Status
Generation	Generation 1	LOCKED
Maximum occupancy	8 persons	LOCKED LIMIT
Planning design mass	28,000 kg	TARGET / OPEN
Main rotor diameter	33.0 m	TARGET / OPEN
Tail rotor diameter	4.0 m	TARGET / OPEN
Capsule mass allocation	7,000 kg	PLANNING / OPEN
Landing supports	8 fixed energy-absorbing supports	CONCEPT / OPEN
Recovery	3 primary + 2 reserve + Rev26 attitude/stabilization enhancements	STUDY / OPEN
Occupant protection	16 airbags + crashworthy seats + F14 adaptive pulse	TARGET / STUDY
Capsule control	F01 + F02 + F03	CONCEPT / OPEN
Water survival	F08 flotation / sealing / stability	TARGET / OPEN
Fire isolation	F09 survival-critical fire isolation	CONCEPT / OPEN
Structural monitoring	F15 capsule/support monitoring	CONCEPT / OPEN
Emergency communications/data	F16/F17/F18	TARGET / OPEN
Core separation	Selective, staged, non-rotor only	CONCEPT / OPEN
Main-rotor separation	Excluded from core safety case	DEFERRED — RESEARCH
Safety authorization	Deterministic Safety Kernel	CONCEPT / OPEN
Digital engineering	F20 6-DOF digital twin / simulation framework	PROGRAMME / OPEN
Verification architecture	F21 demonstrator/test architecture	PROGRAMME / OPEN

10. REV26 OPEN ITEMS AND NEW CLOSURE REQUIREMENTS

ID	Rev26 closure item	Priority
R26-OI-01	33.0 m rotor structural design, blade/hub BoM, dynamic clearance and ground resonance	P0
R26-OI-02	28,000 kg bottom-up mass, CG, inertia and uncertainty closure	P0
R26-OI-03	F01 micro-jet authority, propellant, thermal/fire safety and 6-DOF controllability	P0
R26-OI-04	F06/F07 recovery stabilization and canopy deployment envelope	P0
R26-OI-05	F08 flotation, sealing, water-entry and post-ditching stability	P1
R26-OI-06	F09 fire-isolated capsule thermal/fire containment	P1
R26-OI-07	F10/F11 smart release and partial/asymmetric separation validation	P0
R26-OI-08	F12 terrain-relative recovery logic and validated sensing envelope	P1
R26-OI-09	F13 self-righting mechanism dynamics and packaging	P1
R26-OI-10	F14 adaptive crash-pulse load-path and occupant injury analysis	P0
R26-OI-11	F15 structural health monitoring coverage and false-positive/negative analysis	P1
R26-OI-12	F16/F17/F18 independent post-event power/data/comms closure	P1
R26-OI-13	F19/F20 formal state-machine and 6-DOF model requirements	P0
R26-OI-14	F21 integrated demonstrator and instrumentation plan	P0

All Rev25 OI-01 through OI-18 and R24-OI-01 through R24-OI-05 remain part of the controlled baseline. Rev26 adds, rather than hides, the new closure burden introduced by the expanded architecture.

11. REV26 DOCUMENT-CONTROL AND SOURCE-PRESERVATION STATEMENT

The complete Rev25 Professional Master Edition is preserved after this Rev26 change set so that no original engineering text, table, open item, terminology, or prior-art posture is silently discarded. The Rev26 pages are controlling for the 33.0 m / 28,000 kg configuration and the 21 added features.

Controlled hierarchy: Rev26 configuration statements → Rev26 mass/power calculations → Rev26 feature definitions → retained Rev25 baseline where not explicitly changed → subordinate engineering documents and future evidence.

Configuration-control trigger: changes to occupancy, design mass, rotor geometry, capsule mass, feature architecture, separation logic, recovery architecture, landing-support geometry, airbag architecture, seat stroke or Safety Kernel preconditions require formal impact assessment and document-tree reconciliation.

Integrity statement: numerical values are TARGET, ASSUMPTION, STUDY or CALC unless separately evidenced. This revision is not certification approval, flight-test evidence, a patent grant, a novelty/FTO determination, or demonstrated occupant-survival performance.

END OF REV26 INTEGRATED CHANGE SET

The following pages reproduce the complete Rev25 controlled source content for traceability and preservation.

HSH HEAVY SAFE HELICOPTER

HSH-00 Rev 26 — Integrated Professional Master Edition

ANNEX A — MAIN-ROTOR SEPARATION RESEARCH AND VERIFICATION TRACK

Controlled conceptual extension appended after the Rev 26 baseline

Purpose. This annex records a research-track architecture for the possible complete separation of the 33.0 m main rotor system before recovery-canopy deployment. It does **not** convert rotor separation into the Rev 26 core safety case. It establishes the logic, state sequence, analysis gates, evidence requirements, and failure-containment philosophy that must be satisfied before any future configuration could claim credit for rotor separation.

Baseline relationship. Pages 1–64 of this document remain the controlled Rev 26 baseline. The existing baseline explicitly excludes main-rotor separation from the core safety case and classifies it as deferred research. This annex therefore supplements rather than silently replaces that decision.

Engineering status. All values, timings, mechanisms, trajectories, structural responses, and performance outcomes in this annex are conceptual unless explicitly supported by analysis or test. No detailed release mechanism geometry, actuator sizing, energetic-device specification, or flight procedure is defined here.

Safety principle. The desired sequence is not “separate quickly and deploy a parachute after a fixed delay.” The intended logic is: emergency confirmation → controlled rotor safe-state assessment → positive separation authorization → verified divergence/clearance → recovery authorization inside a validated envelope.

Document control

Field	Value
Document	HSH-00 Rev 26 — Annex A
Baseline	HSH-00 Rev 26, 33.0 m main rotor / 28,000 kg planning configuration
Annex status	CONCEPT / RESEARCH TRACK / OPEN
Core safety case	Unchanged; main-rotor separation remains excluded
Primary purpose	Define a verification-led path for possible future inclusion
Certification status	Not certified; no airworthiness credit implied
Flight status	Not flight-tested

A1. CONTINUITY WITH THE REV 26 SAFETY CASE

Rev 26 establishes a layered survival architecture based on conventional recovery, autorotation where available, selective non-rotor separation, capsule release, recovery canopies, impact attenuation, and post-landing survival. The existing document deliberately assumes that the main rotor remains attached.

The present annex preserves that assumption as the **fallback baseline**. Rotor separation is treated as an optional research capability that may only become a credited path after independent evidence demonstrates that the rotor can depart the occupied capsule's hazard envelope without creating a larger hazard.

Existing Rev 26 position	Annex A treatment
Main rotor separation excluded from core safety case	Retained unchanged
Rotor separation listed as deferred research	Expanded into a structured research and verification programme
Capsule clearance required before recovery	Extended to include rotor-disk and rotor-body clearance
S5 recovery authorization is not time-gated	Retained; clearance and validated envelope remain mandatory
AI/ML advisory only	Retained; deterministic Safety Kernel remains final authorization authority
Human emergency authority non-bypassable	Retained

A1.1 Non-regression rule

No future rotor-separation development shall weaken the existing no-rotor-separation survival path. If a rotor-separation feature is unavailable, inhibited, partially executed, or judged outside its validated envelope, the system shall revert to a previously validated survival mode or to the best available degraded state. A research feature shall never become a single point of failure for the baseline survival architecture.

A2. PROPOSED FUNCTIONAL SEQUENCE

The conceptual sequence below replaces the earlier assumption that a rotor could simply be released and followed by a fixed parachute delay. The critical transition is not elapsed time; it is verified physical state.

State	Function	Entry condition	Exit gate
R0	Emergency detection / confirmation	Validated catastrophic condition	Independent confirmation
R1	Rotor safe-state assessment	Emergency confirmed	Rotor/engine/attitude state within validated preconditions or defined degradation
R2	Energy and hazard isolation	Separation authorized for preparation	Required energy/fuel/fire isolation confirmed
R3	Rotor separation authorization	All mandatory preconditions true	Deterministic authorization + human authority path available
R4	Positive rotor separation	Authorized release state	Positive release-state confirmation
R5	Divergence / clearance verification	Rotor no longer structurally retained	Rotor and capsule trajectories inside validated clearance envelope
R6	Capsule recovery authorization	Clearance verified	Recovery envelope valid
R7	Canopy deployment	Recovery authorized	Canopy state confirmed
R8	Descent / landing survival	Recovery established	Ground contact / post-landing state

The key design change is the insertion of **R4 Positive Rotor Separation** and **R5 Divergence/Clearance Verification** as explicit states. R6 must not be triggered solely by a timer.

A2.1 Timing philosophy

A short sequence is desirable only insofar as it is physically achievable and verified. A nominal “few seconds” target may be retained as a programme objective, but the safety logic shall not use elapsed time as proof of separation or clearance.

A3. ROTOR-SPECIFIC SAFETY KERNEL EXTENSION

The existing Safety Kernel precondition set already covers altitude/airspeed, attitude/angular rates, rotor and engine state, structural/attachment health, trajectory/clearance, recovery integrity, power/communications, and human authority. Rotor separation requires these conditions to be made more explicit for the rotor itself.

Precondition	Required evidence before authorization
Rotor state known	Independent and valid rotor-state sensing; validity monitoring
Retention state known	Positive indication of the structural attachment state
Energy state bounded	Rotor rotational/drive state inside the validated separation envelope
Attitude/rates bounded	Vehicle angular state within the validated separation envelope
Capsule path valid	Predicted capsule trajectory remains inside the recovery corridor
Rotor path valid	Predicted rotor trajectory does not intersect capsule or recovery system
Relative clearance valid	Minimum separation envelope demonstrated across uncertainties
Release symmetry valid	No unresolved partial/asymmetric release condition
Recovery system valid	Canopy deployment remains inside validated aerodynamic/dynamic envelope
Fallback available	A defined response exists for every detected separation anomaly
Human authority valid	Crew inhibit/command path remains available and deterministic

A3.1 Hard inhibit concept

The Kernel shall inhibit recovery deployment whenever rotor/capsule clearance is unknown, unresolved, or outside the validated envelope. The same principle applies if separation-state sensing disagrees across independent channels. The correct response to uncertainty is a defined degraded state—not an assumption that the rotor has cleared.

A3.2 Sensor disagreement

Sensor disagreement is a first-class state. The design shall specify how disagreement is detected, latched, reported, and handled without allowing a single failed or implausible indication to authorize canopy deployment.

A4. FAILURE-CONTAINMENT ARCHITECTURE

Failure / hazard	Consequence	Required containment concept	Evidence
No release	Rotor remains attached	Revert to validated no-rotor-separation path	System simulation + ground test
Partial release	Asymmetric forces / uncontrolled rotor motion	Immediate state identification; prohibit recovery until validated state transition	Validated state transition test
Delayed release	Reduced time margin	Time is advisory only; use state-based gates	Timing uncertainty analysis
Unexpected rotor trajectory	Capsule collision risk	Clearance inhibit; deterministic fallback	Monte-Carlo / uncertainty envelope
Rotor contacts recovery system	Canopy damage / loss of recovery	Trajectory separation requirement	Dynamic interaction analysis
Loss of release-state sensing	Unknown physical state	No recovery authorization until alternate evidence or validated fallback	Validated fallback
Loss of power	Actuation/sensing degradation	Independent survival power and safe-state logic	Power-loss test
Control degradation	Attitude divergence	Defined limits and degraded recovery mode	6-DOF simulation
Unexpected structural response	Attachment failure / debris	Structural qualification and exclusion zones	FEA + subscale/full-scale test
Canopy outside envelope	Inflation instability / failure	Hard deployment inhibit	Canopy deployment test

The central containment rule is that **rotor separation failure must not become capsule recovery failure**. The architecture therefore separates the decision to attempt rotor release from the decision to authorize parachute deployment.

A5. 6-DOF DYNAMICS AND TRAJECTORY VERIFICATION

A credible rotor-separation case requires a six-degree-of-freedom dynamic model of the aircraft/capsule and a corresponding model of the separated rotor assembly. The model must represent translation, rotation, inertia, aerodynamic forces and moments, relevant actuator/drive states, release-state uncertainty, and environmental variation at the level necessary to establish a defensible clearance envelope.

The objective is not to produce a single nominal trajectory. The objective is to determine whether the capsule and rotor remain separated across credible variations in initial state, structural tolerances, sensor uncertainty, timing dispersion, attitude/rate conditions, and aerodynamic environment.

Analysis block	Minimum output
Rigid-body 6-DOF	Time history of position, velocity, attitude and angular rates
Rotor rigid-body model	Rotor position/orientation and inertial response after release
Attachment/release model	State transition and uncertainty bounds at release
Aerodynamic interaction	Forces/moments and wake/interference sensitivity
Initial-condition sweep	Validated operating envelope rather than one nominal case
Uncertainty propagation	Clearance margins under parameter uncertainty
Collision geometry	Capsule/rotor/line/canopy exclusion volumes
Sensitivity analysis	Identification of dominant drivers of clearance
Degraded cases	Partial/asymmetric/late-release trajectories

A5.1 Verification criterion concept

A future engineering requirement should be expressed as a bounded clearance condition over the validated uncertainty set, not as a statement such as “the rotor will fall away.” The final numerical margin must come from analysis and test and is intentionally not assigned in this conceptual annex.

A6. STRUCTURAL AND ROTOR DYNAMICS VERIFICATION

The rotor system is a large rotating structure. The separation case therefore requires structural evidence covering the rotor, hub, attachment load paths, surrounding airframe structure, and the transient response created by the change from an attached rotating system to a released body.

Discipline	Question to close	Evidence path
Rotor structural dynamics	Does the rotor remain structurally controlled through the separation event?	FE validated dynamic model / test
Hub and attachment	Can the attachment path transition without unacceptable failure modes?	Validated structural analysis + component test
Airframe response	What transient loads reach the capsule/airframe?	Coupled structural analysis + test
Gyroscopic response	What attitude/moment effects occur during release?	Rigid-body + flexible-body simulation
Ground resonance / coupled modes	Are there relevant pre-release dynamic states?	Rotor dynamic analysis
Blade deformation	Can elastic motion change the clearance envelope?	Aeroelastic / structural dynamics
Fragment/debris control	Can structural failure create an additional hazard?	Failure analysis + containment assessment

The analysis must include not only nominal separation but credible non-nominal states. The existing Rev 26 architecture already flags rotor dynamics and clearance as open. Annex A turns that open item into a defined evidence stream rather than declaring the capability solved.

A7. CAPSULE-ROTOR-CANOPY CLEARANCE ARCHITECTURE

Clearance is the central interface between rotor separation and parachute recovery. Three moving systems must be considered: the released rotor assembly, the descending capsule, and the recovery canopy/riser/line system. The envelope must also account for structural deformation and uncertainty.

Object pair	Required check
Rotor ↔ capsule	No damaging intersection throughout the validated transition
Rotor ↔ canopy	No rotor encounter with canopy, risers, or recovery lines
Rotor ↔ deployment flow	No unacceptable aerodynamic interaction during canopy inflation
Capsule ↔ rotor wake	Assess dynamic effects relevant to attitude and canopy deployment
Capsule ↔ recovery hardware	No line/attachment geometry conflict
Partial-release rotor ↔ capsule	Explicit degraded trajectory and inhibit logic
Rotor ↔ ground	Ground-interaction consequences considered where relevant

A7.1 Clearance verification sequence

The preferred logic is: detect release-state transition → establish relative state estimate → compare against validated envelope → continue monitoring → authorize recovery only when the required clearance conditions are positively established. A simple elapsed-time delay is insufficient because timing does not establish spatial separation.

A8. RECOVERY-CANOPY INTEGRATION

Rev 26 defines a 3-primary + 2-reserve recovery architecture and notes that minimum deployment altitude is an open critical item. Annex A does not change that architecture. Instead, it adds the requirement that rotor separation be treated as a dynamic precursor whose outcome must be included in canopy deployment validation.

Recovery condition	Authorization principle
Rotor status unknown	Inhibit canopy deployment
Rotor not positively clear	Inhibit canopy deployment
Rotor clear but capsule state outside canopy envelope	Inhibit or enter validated degraded recovery
Rotor clear + capsule state valid	Recovery authorization may proceed
Primary canopy failure	Reserve logic per validated recovery architecture
Low-altitude condition	Use only procedures/branches validated for that altitude regime

The annex intentionally does not set a universal “seconds after rotor release” value. The correct deployment point depends on the validated dynamic envelope, minimum altitude, attitude/rate limits, canopy inflation characteristics, and the actual separation trajectory.

A9. STATE MACHINE — EXPANDED VERSION

State	Description	Key gate
NORMAL	Normal flight	All locks and monitoring nominal
ANOMALY	Abnormal condition detected	Independent confirmation
ASSESSMENT	Impact/severity/recovery assessment	Safety Kernel preconditions
RECOVERY ATTEMPT	Conventional control / OEI / autorotation	Success or failure
ROTOR PREPARE	Prepare rotor research track	Rotor state, energy, structure, attitude valid
ROTOR AUTHORIZE	Authorize rotor separation	All mandatory conditions simultaneously true
ROTOR RELEASE	Execute controlled separation	Positive release-state confirmation
ROTOR DIVERGENCE	Monitor relative trajectories	Clearance envelope
CAPSULE CLEAR	Capsule path verified	Recovery envelope valid
RECOVERY AUTHORIZE	Permit canopy deployment	Deterministic authorization
DESCENT	Canopy flight	Attitude/rate/canopy monitoring
LANDING	Impact attenuation	Supports/seats/airbags per existing architecture
POST-LANDING	Survival and rescue	Locator/comms/life support
AUDIT	Preserve event data	Data secured

A9.1 State-machine rule

Every transition into rotor separation shall be reversible or fail-safe until the physical separation event has actually occurred. Once release begins, the system shall treat partial/asymmetric separation as a distinct state rather than forcing the event into a nominal-success branch.

A10. VERIFICATION AND VALIDATION GATES

Gate	Evidence required	Exit condition
G0 — Requirements	Requirements, hazards, interfaces	Approved baseline
G1 — Modeling	6-DOF + structural + clearance models	Model credibility established
G2 — Component	Attachment/release component tests	Required failure modes characterized
G3 — Integrated ground	Integrated separation and sensing tests	State transitions demonstrated
G4 — Dynamic validation	Representative dynamic/rig testing	Model-to-test correlation
G5 — Recovery integration	Capsule + canopy + clearance integration	Deployment envelope validated
G6 — Environmental	Relevant temperature/vibration/power/EMI cases	Environmental robustness shown
G7 — Flight research	Controlled flight-test programme under approved basis	Evidence supports candidate operational envelope
G8 — Configuration credit	Safety/certification review and traceability	Formal configuration decision

No gate is passed by documentation alone. Each gate requires objective evidence appropriate to the claim. A successful ground release test does not by itself prove safe free-flight trajectory or parachute clearance.

A11. TEST PROGRAMME STRUCTURE

Test level	Purpose	Representative output
Material / subcomponent	Characterize structural and interface behavior	Load/deflection/failure data
Release component	Demonstrate controlled state transition	Release-state repeatability
Rotor/hub representative	Characterize transient response	Dynamic response data
Integrated ground article	Verify sequencing and sensing	State-machine evidence
Dynamic rig	Validate 6-DOF model assumptions	Model correlation
Recovery integration	Validate clearance and deployment interaction	Clearance/deployment evidence
Environmental	Verify robustness of sensors/logic/power	Fault and environment data
Flight research	Validate real aerodynamic and dynamic behavior	Approved flight-test evidence

A11.1 Test philosophy

Testing shall proceed from low-consequence, highly instrumented configurations toward higher-fidelity configurations. The programme should deliberately expose failure modes rather than testing only nominal releases. Particular attention is required for partial release, delayed release, sensor disagreement, power interruption, and trajectory uncertainty.

No test article should be interpreted as evidence of production readiness unless the configuration, loads, instrumentation, uncertainty, and acceptance criteria are traceable to the configuration under review.

A12. REQUIREMENTS TRACEABILITY MATRIX

Requirement ID	Requirement statement	Verification
RSA-001	Rotor separation shall remain outside the core safety case until evidence supports configuration credit.	Review
RSA-002	Recovery authorization shall be deterministic and state-based.	Analysis + test
RSA-003	Rotor release shall have positive physical-state indication.	Test
RSA-004	Partial/asymmetric release shall be detected and handled.	Fault injection + test
RSA-005	Rotor/capsule clearance shall be demonstrated over a validated uncertainty envelope.	6-DOF analysis + test
RSA-006	Rotor/canopy interaction shall be assessed before recovery credit.	Dynamic analysis + test
RSA-007	Loss of sensing shall not create an automatic recovery authorization.	Fault injection
RSA-008	The existing no-rotor-separation survival path shall remain available.	System integration
RSA-009	Human emergency authority shall remain non-bypassable.	HMI/system test
RSA-010	AI/ML shall not be the final safety authorization authority.	Architecture review
RSA-011	Timing targets shall not replace physical clearance evidence.	Requirements audit
RSA-012	Configuration credit shall require formal change control.	Configuration audit

A13. CONFIGURATION AND MASS IMPACTS

Rev 26 allocates approximately 3,680 kg to the main rotor/hub/controls and 1,000 kg to the rapid separation system. The rotor-separation research track may affect structure, sensors, release hardware, reinforcement, power, control logic, wiring, verification instrumentation, and configuration reserve. No additional mass is declared closed by this annex.

Configuration item	Rev 26 baseline	Annex A treatment
Main rotor / hub / controls	≈3,680 kg planning	Retain until detailed design/analysis
Rapid separation system	1,000 kg planning	Existing non-rotor allocation; rotor-specific increment OPEN
Capsule	7,000 kg planning	Unchanged in concept; interfaces require reassessment
Recovery system	3 primary + 2 reserve	Architecture retained; dynamic envelope reassessment required
Occupancy	8 maximum	Unchanged
Design mass	28,000 kg planning	No closure claimed
Rotor diameter	33.0 m	Unchanged

A13.1 Configuration-control trigger

Any change to rotor mass, hub geometry, attachment structure, release interfaces, capsule attachment points, recovery attachment geometry, or Safety Kernel preconditions shall trigger formal configuration impact analysis across mass properties, CG, inertia, dynamics, structures, controls, recovery, and verification.

A14. RESEARCH RISK REGISTER

Risk	Status	Principal closure activity
Rotor release produces unsafe attitude response	OPEN — CRITICAL	6-DOF coupled dynamics + test
Rotor remains partially attached	OPEN — CRITICAL	Positive state sensing + asymmetric-load analysis
Rotor intersects capsule	OPEN — CRITICAL	Validated clearance envelope
Rotor intersects canopy	OPEN — CRITICAL	Canopy interaction analysis
Release timing variability reduces margin	OPEN	Timing uncertainty characterization
Structural transient exceeds assumptions	OPEN	Coupled structural dynamics + test
Recovery envelope becomes invalid after release	OPEN — CRITICAL	Integrated recovery analysis
Sensor disagreement falsely indicates clearance	OPEN — CRITICAL	Diverse sensing + fault injection
Loss of power during transition	OPEN	Independent power / safe-state analysis
Rotor separation becomes mandatory for safety	PROHIBITED DESIGN DRIFT	Configuration governance
Insufficient evidence for certification credit	OPEN	Evidence ladder + formal review

A14.1 Design-governance rule

The research programme shall not use success bias. A nominally successful separation demonstration cannot close hazards that were not represented in the test or analysis. The evidence record must state the tested envelope, excluded cases, uncertainty, and remaining limitations.

A15. INTEGRATED CONCLUSION AND CHANGE-CONTROL STATEMENT

Annex A defines a credible engineering path for investigating complete main-rotor separation while preserving the central safety philosophy of HSH-00 Rev 26. The essential architectural decision is to treat rotor separation as a **research-track capability**, not as an assumption that can be inserted into the existing survival chain merely by adding a short timer.

The proposed sequence is therefore: **confirm emergency → assess rotor state → isolate hazards → authorize separation → positively verify release → verify rotor/capsule divergence and clearance → authorize recovery → deploy canopies within a validated envelope → continue existing landing and post-landing survival architecture.**

The existing Rev 26 fallback remains intact. If rotor separation cannot be demonstrated, the aircraft does not lose the survival logic already defined around the attached-rotor assumption.

The principal closure work is six-degree-of-freedom dynamics, structural and rotor transient analysis, clearance-envelope definition, sensing and state verification, integrated ground testing, recovery-system interaction testing, and ultimately a controlled flight-research programme under an appropriate certification and safety framework.

No exact release mechanism, energetic device, actuator sizing, cut-point geometry, or flight-operational procedure is specified in this annex. Those details require detailed structural, systems-safety, certification, supplier, and test engineering.

Configuration status

Item	Annex A disposition
Main-rotor separation in core safety case	NO — remains excluded
Research architecture defined	YES — conceptual
6-DOF evidence	OPEN
Structural evidence	OPEN
Clearance evidence	OPEN — CRITICAL
Mechanism qualification	OPEN
Recovery integration evidence	OPEN
Flight evidence	OPEN
Certification credit	NONE

END OF ANNEX A — HSH-00 REV 26

Controlled conceptual material. Not certified. Not flight-tested. Not a filed patent. All future changes require formal configuration control.